02 Cell based Design Concept

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Course Outline

Impacts from VLSI Technology Migration

- System-on-a-Chip (SoC) and System-in-a-Package (SIP)
- SoC Design Challenges and Design Issues
- VLSI Design Process: Dealing with Complexity

Introduction to Cell-based Design Flow

- Cell-based Design Flow
- Full-Custom Design Flow
- Wiring Problem and Timing/SI Sign-off

Introduction to Standard Cell-library

- Cell Library Data Sheet
- Cell Library Contents
- Timing Check of Sequential Cells
- Cell Library Layout and Characterization

Design Trend – SoC and SIP

- **Integration improves the design:**
  - lower parasitics = higher speed;
  - lower power;
  - physically smaller.
- **Integration reduces manufacturing cost-(almost) no manual assembly.**

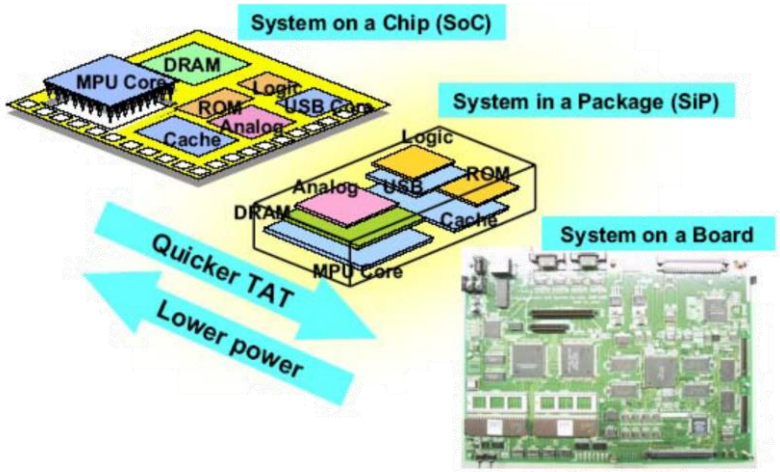


Figure 1.2.6: SiP will be an assembly style suitable for power-aware systems providing low-power and quick TAT feature at the same time.

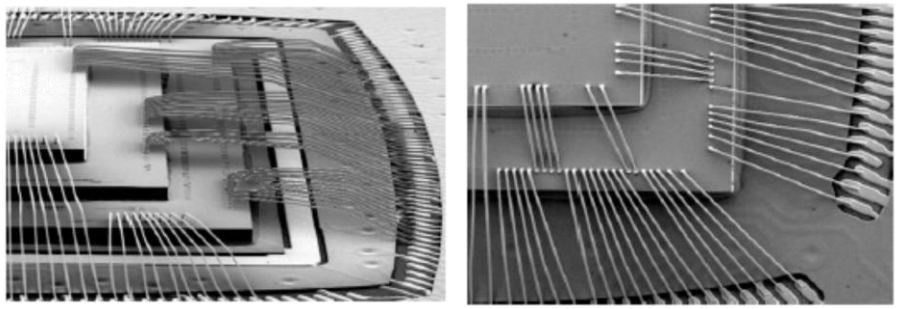


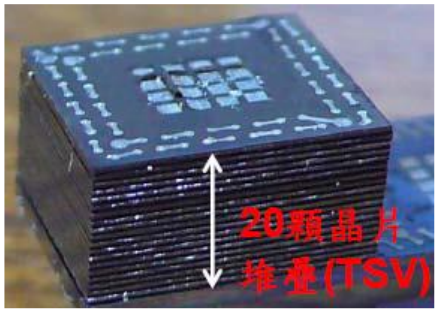
Figure 1.3.12: System-in-Package (SiP): stacked chip modules (SCM).

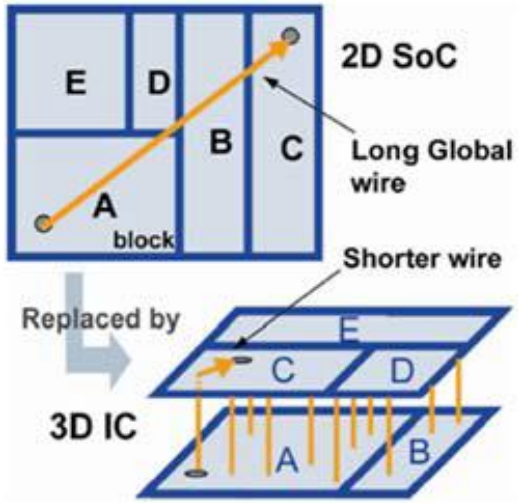
Ref. Plenary 1.2, *2003 ISSCC*

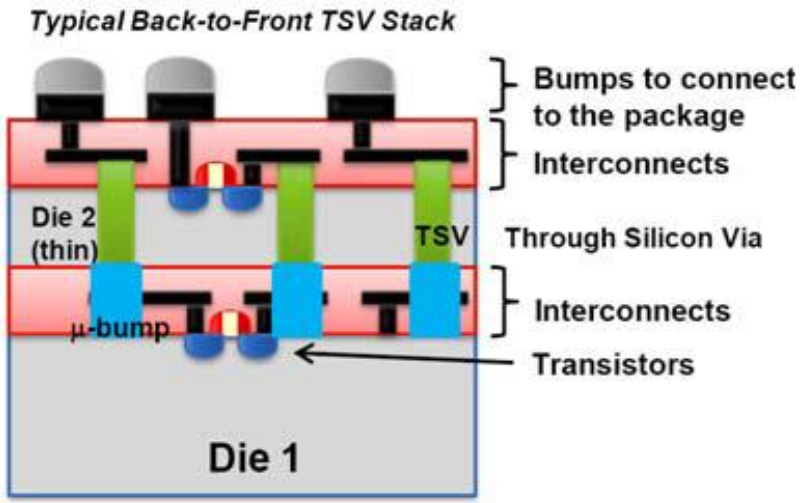
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Design Trend – 3D IC

- **3D IC technology can improve interconnect density.**
  - It reduces parasitic RLC and thus increases circuit speed.
  - TSV: Through Silicon Via



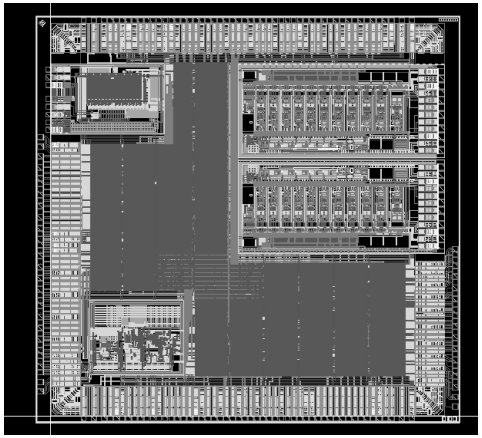




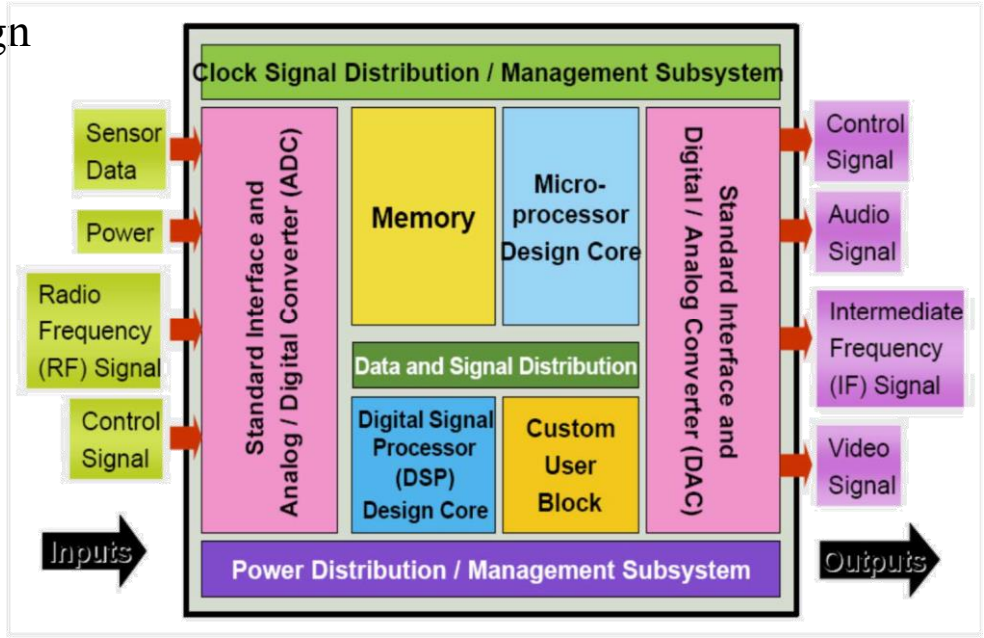
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Design Trend – System-level Integration

- **Toward system-level integration**
  - Provide (multi-)system-on-a-chip (SOC) solution
  - IP-based design
  - Platform-based system design
  - HW/SW co-design



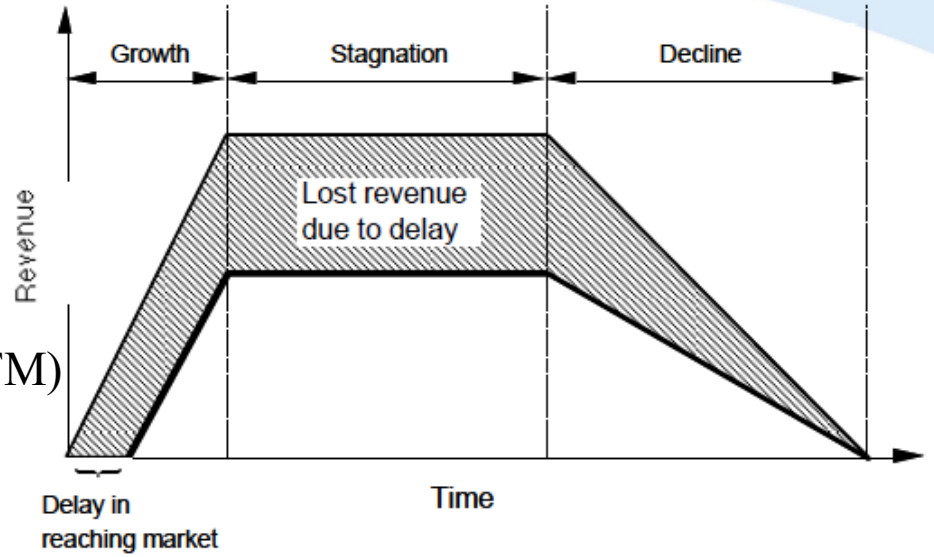
802.11b Baseband



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Impacts from VLSI Technology Migration

- **Design Trend:**
  - system-level integration
  - single-chip solution (SoC)
  - design for testability (DFT)
  - design for manufacturability (DFM)
  - design for reliability

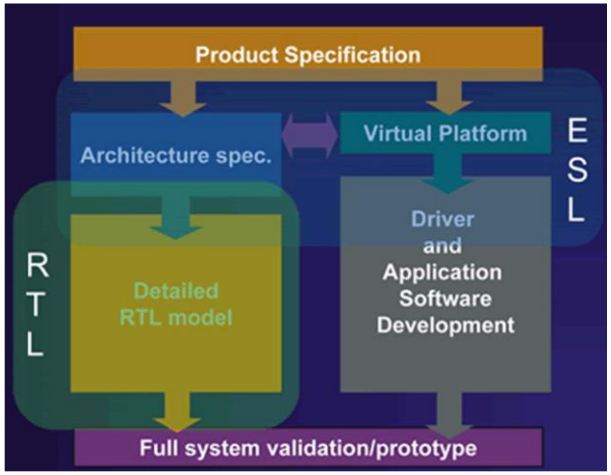


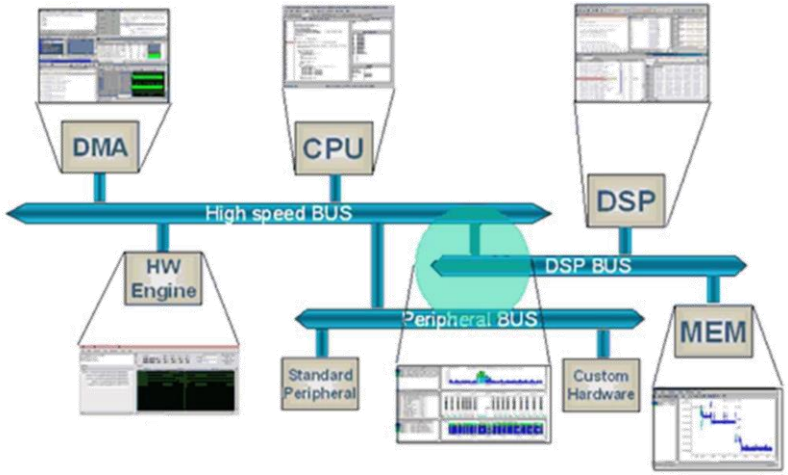
- Resulting in more design complexity and efforts, however design cycle remains the same or even shorter
- New product most profitable period is first 18 months-2 years.
  -1/8 delay of product => lifetime reduces 1/3 revenue.

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Impacts from VLSI Technology Migration (cont'd)

- **CAD Trend:**
  - behavioral synthesis (SystemC-to-RTL or ESL)
  - platform solution (EDA vendor must provide total solution)
  - timing/SI sign-off solution
  - test and verification solution
  - yield improvement chip optimizer

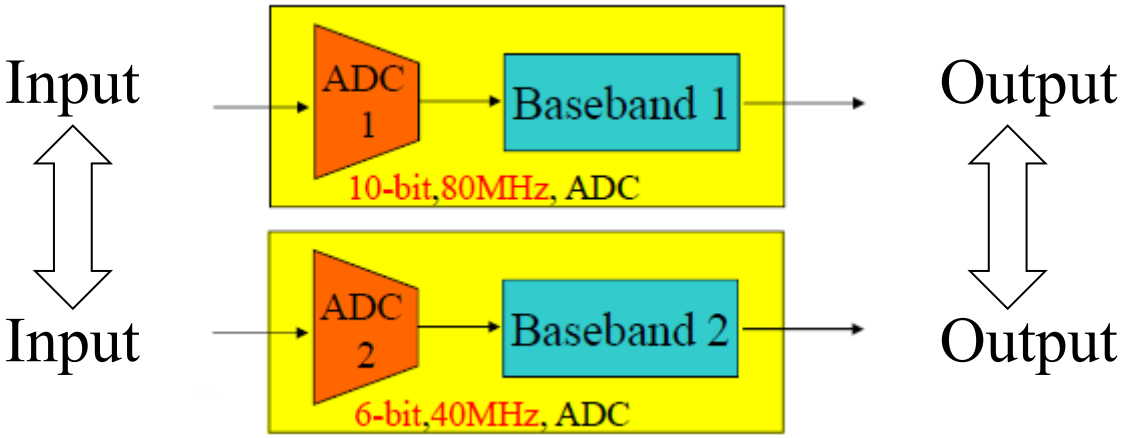




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Toward Cost-effective Realization

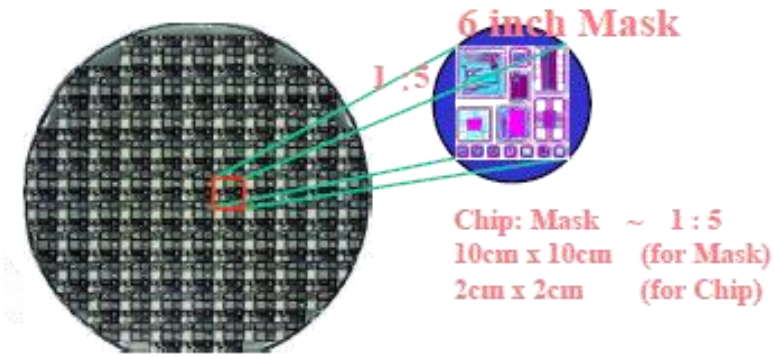
- System-Level Design Considerations lead to more
- cost-effective realization
- IP-based design reduce system turn around time
- IP selection and creation should meet system specs without much interface overhead
- Joint-work of both system and IC designers becomes a must



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Product Design Flow

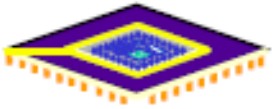
- 1. System Definition
- 2. Architecture selection and circuit simulation
- 3. Layout drawing and verification
- 5. Photo mask and chip fabrication
- 6. Die saw and package
- 7. Chip and system testing
- 8. Yield and testability
- 9. Mass Production





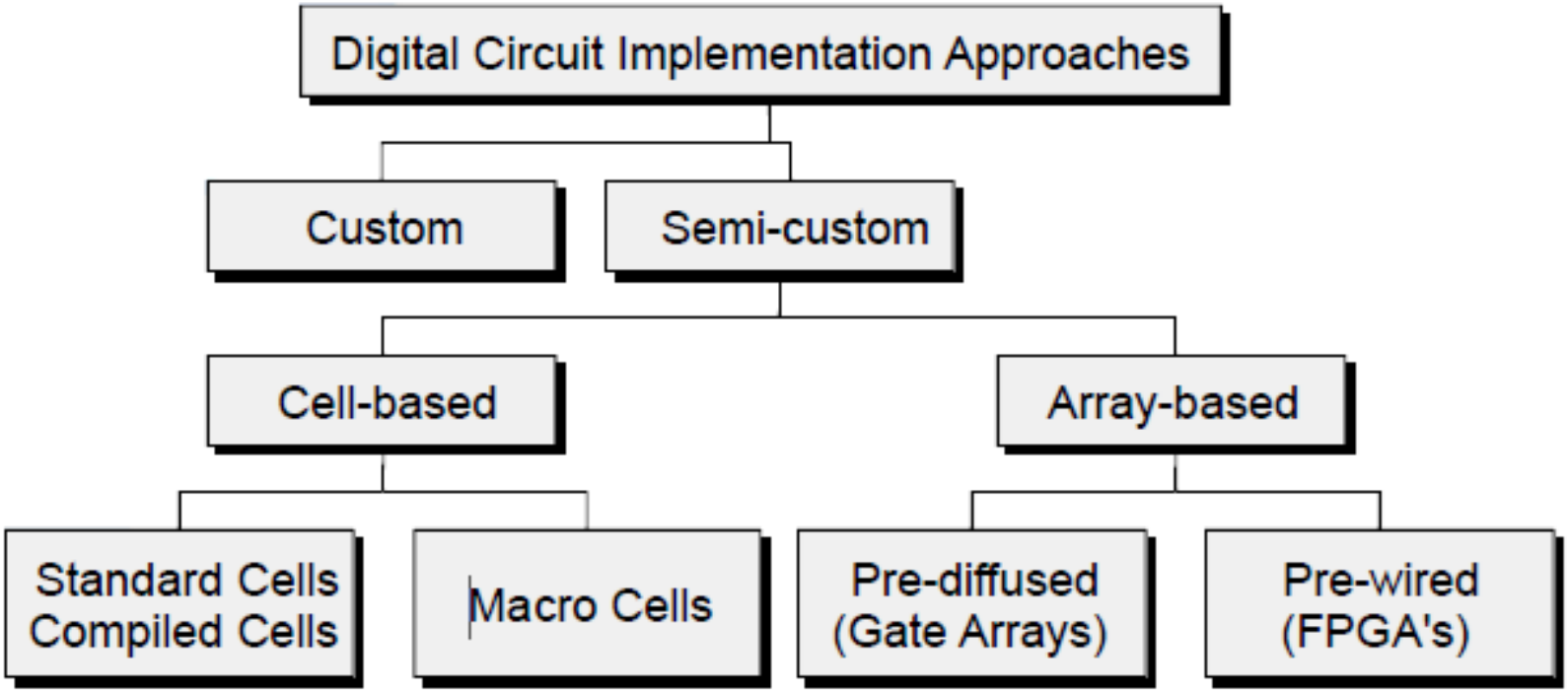






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Implementations Choices



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Challenges in VLSI design

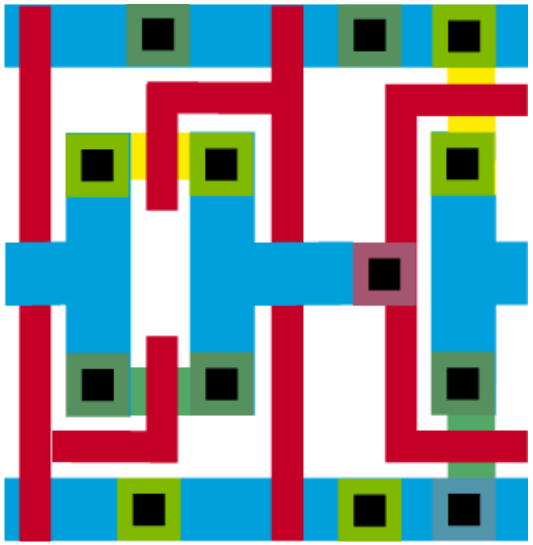
- **Multiple levels of abstraction**
  - transistors to CPUs.
- **Multiple and *conflicting* constraints**
  - low cost and high performance are often at odds.
- **Short design time**
  - Late products are often irrelevant.



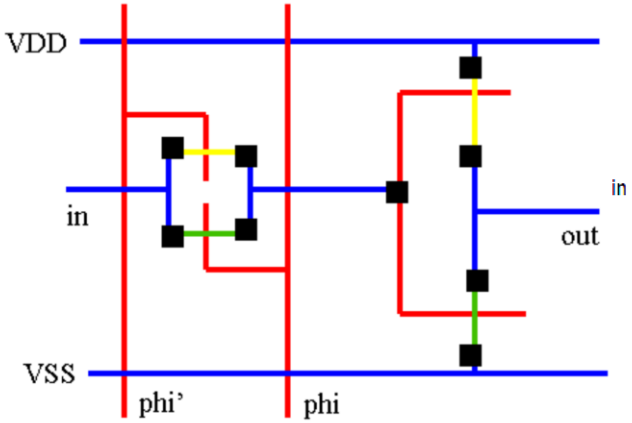
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Layout and its abstractions

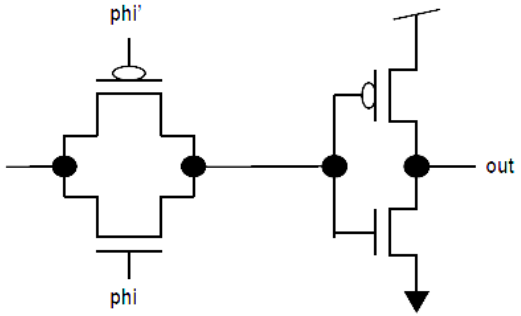
Layout for dynamic latch



Layout for dynamic latch

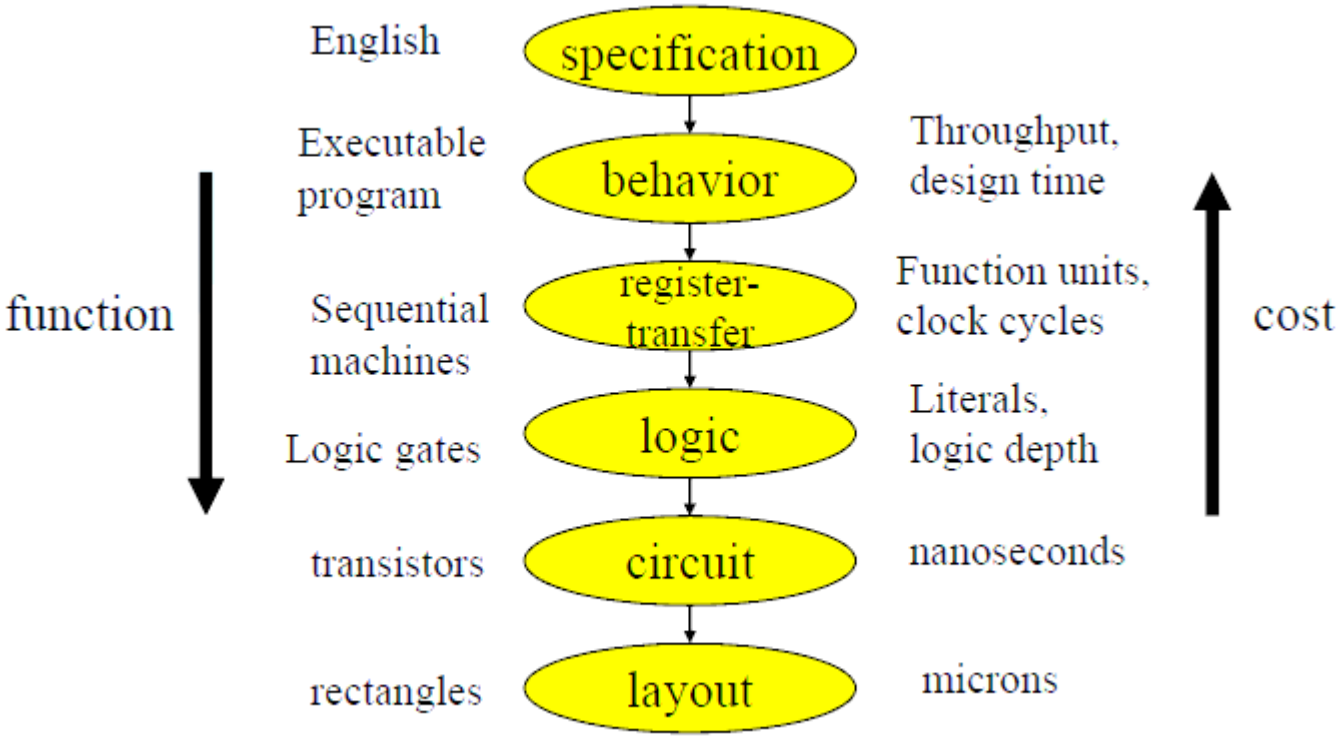


Transistor schematic



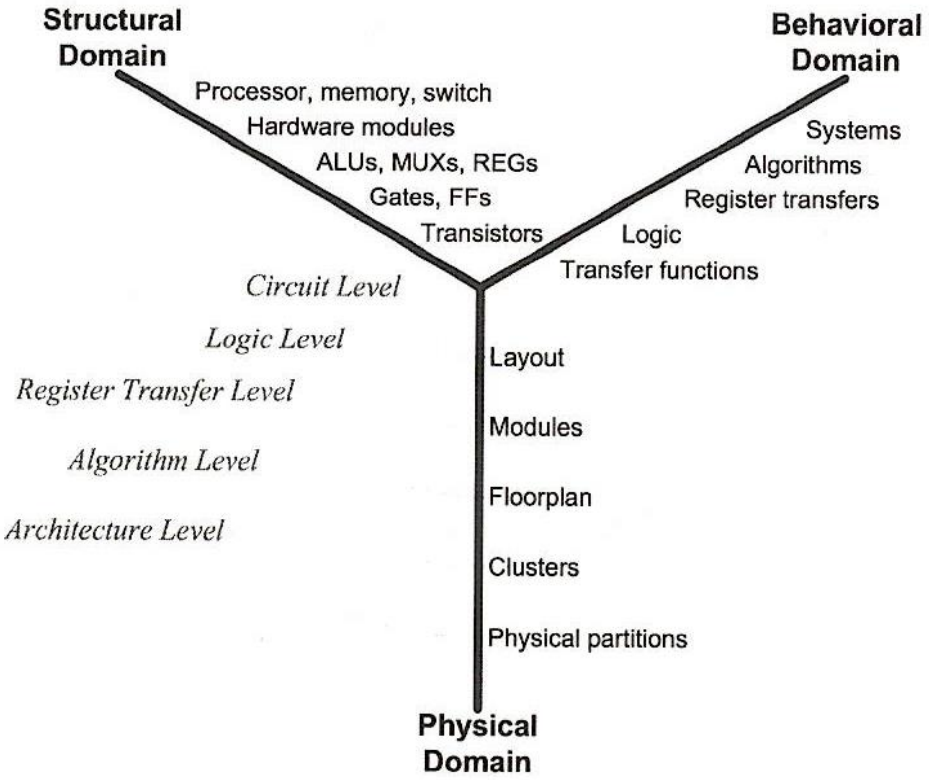
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Design Abstractions (cont'd)



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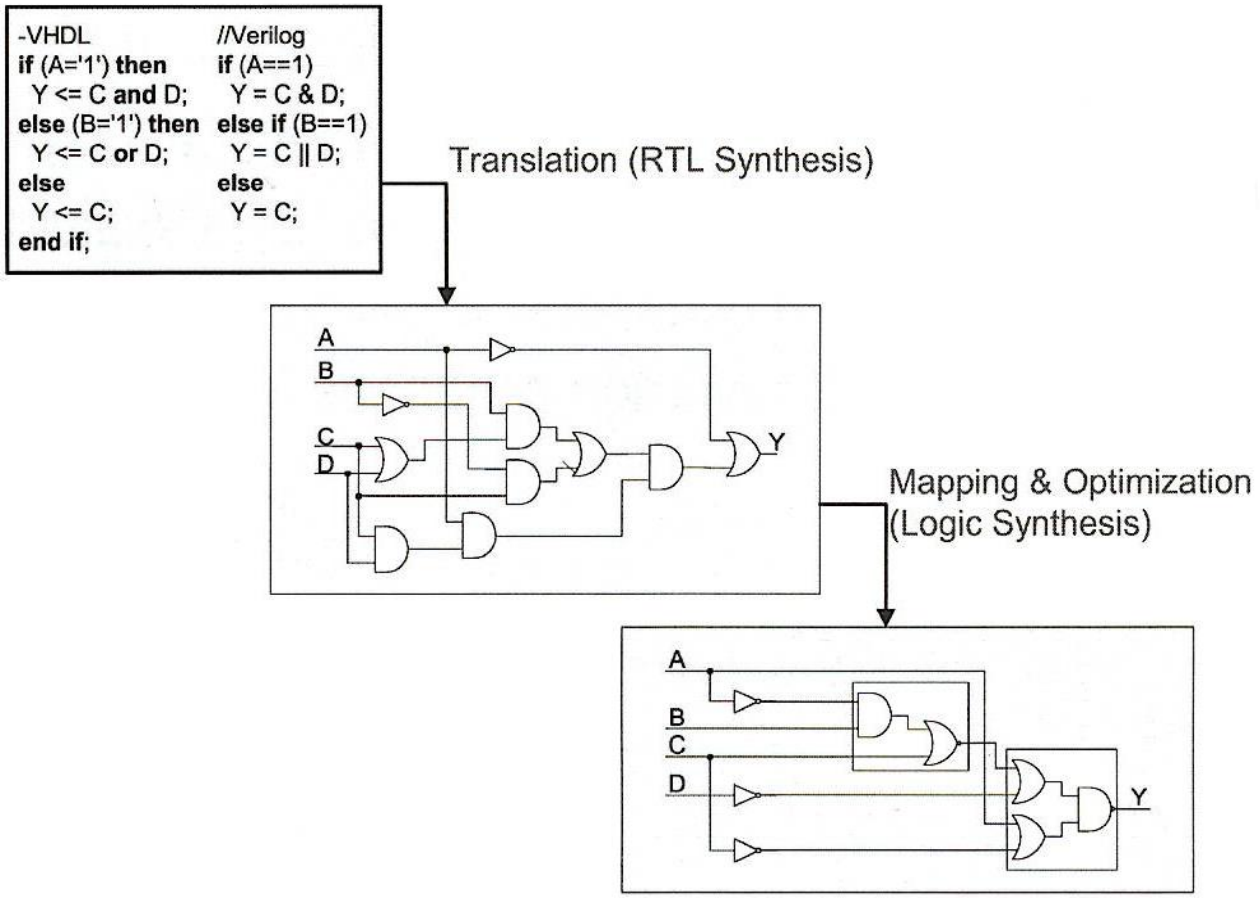
Description Levels



Revised from: Silicon Compilation

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RTL & Logic Synthesis



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RTL & Logic Synthesis

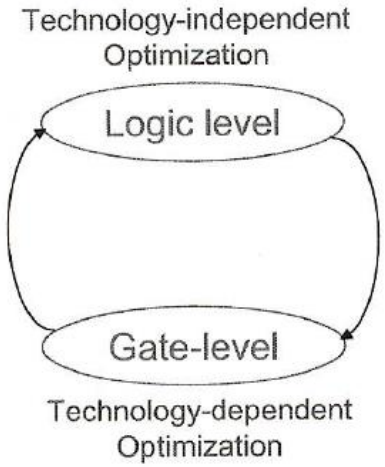
- **Translate HDL code functional equivalent technology dependent gate-level netlist which preserve design requirements**
- **RTL synthesis**
  - HDL code parsing
  - Synthetic library component instantiation
  - RTL optimization
  - Technology independent representation translation
- **Logic synthesis**
  - Logic optimization
  - Technology mapping & optimization

RTL Synthesis Problems

- **How to make the code synthesizable?**
  - Unsupported language constructs
  - Unsupported semantics
- **How to make the before & after synthesis designs consist?**
  - Combinational before , sequential after
  - 8 latches before , 12 latches after
- **How to make the code optimal for synthesis?**
  - Resource sharing

Design Optimization

- **Logic optimization**
  - Flattening
  - Structuring
- **Gate-level optimization**
  - Initial sequential optimization
  - Combination optimization
  - Final sequential optimization
  - Localized adjusting

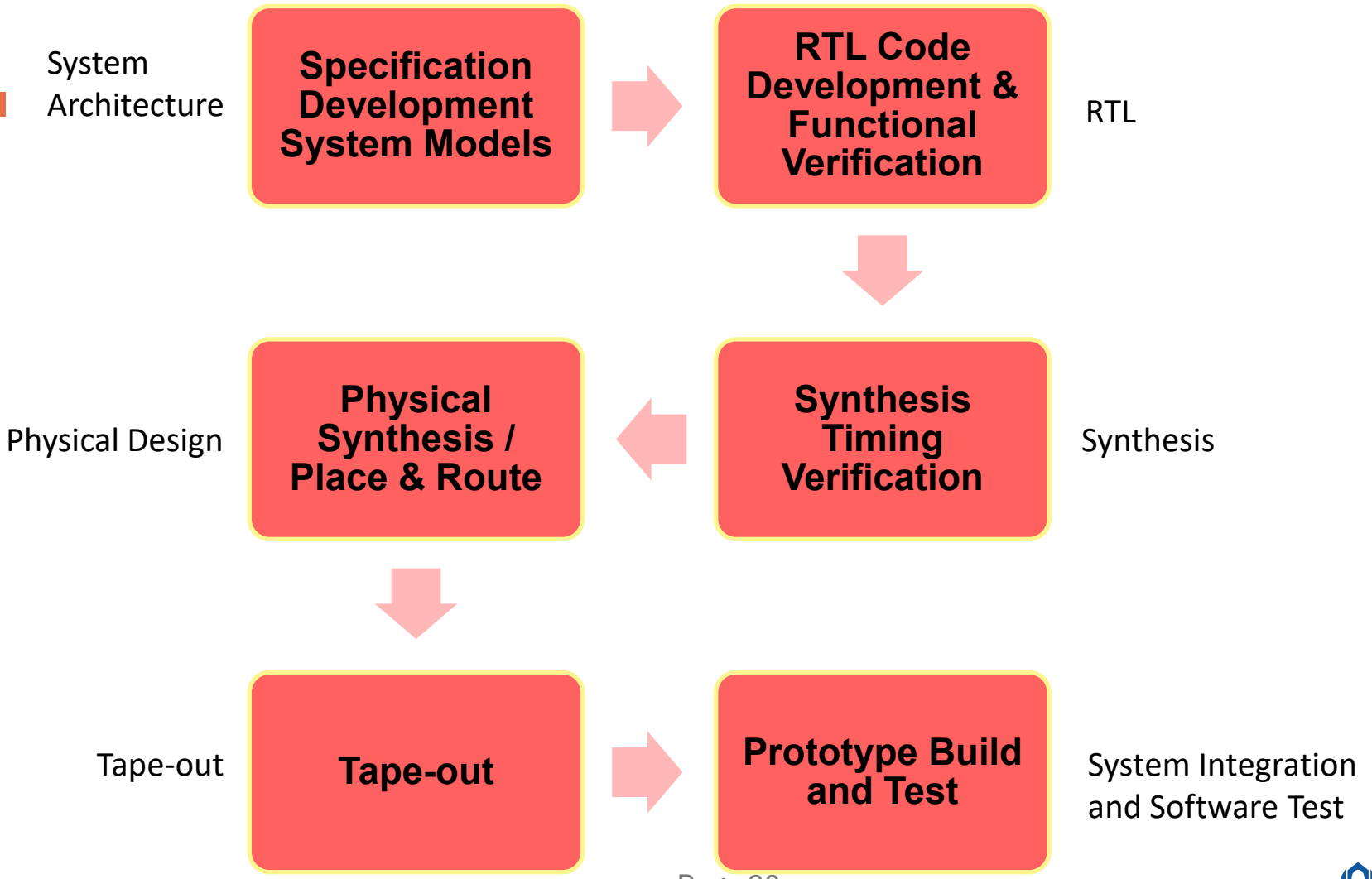


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Cell-based ASIC

- **Cell-based IC (CBIC)**
  - use **pre-designed** logic cells (known as standard cells) and micro cells (e.g. microcontroller)
  - designers save time, money, and reduce risk
  - each standard cell can be optimized individually
  - all mask layers are customized
  - custom blocks can be embedded
  - manufacturing lead time is about 1.5 months

Mature Cell-based Design Flow



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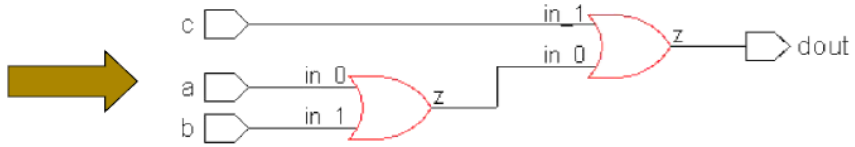
Cell-based Design Flow (cont'd)

- **From Verilog Behavior-Level HDL to Logic Gates(netlist)**

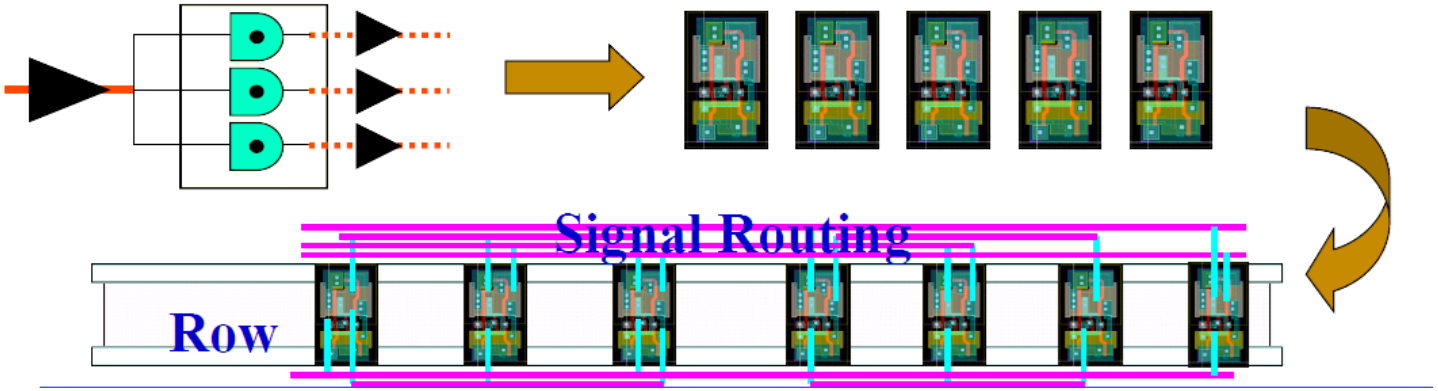
```
module comb_or(dout,a,b,c);
input a,b,c;
output dout;

assign dout = a | b | c;

endmodule
```



- **From Verilog Structural-Level HDL to Layout**



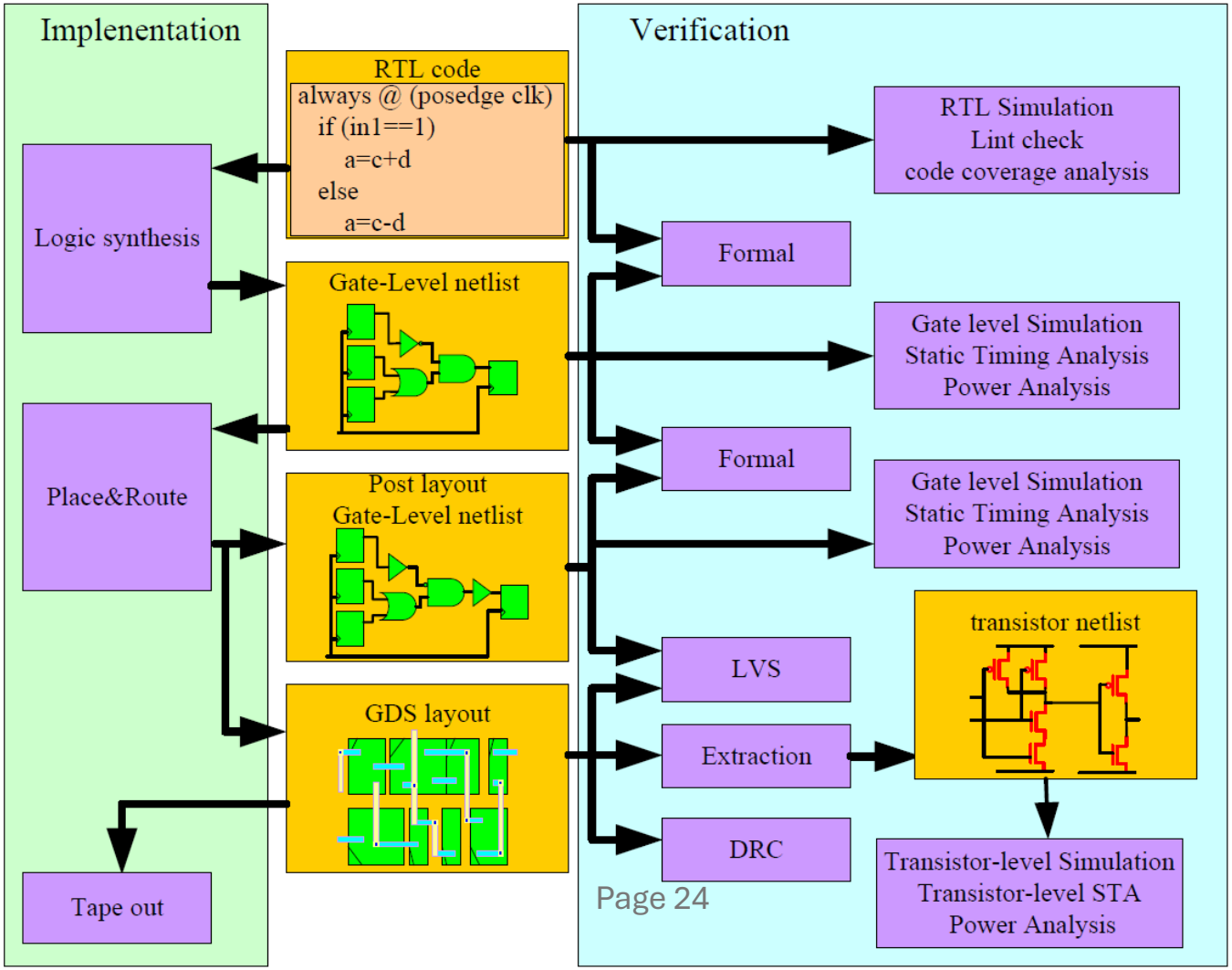
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Cell-based Design Tools

- **System Architecture**
  - C/C++, MATLAB
  - Python, Tensorflow
  - SystemC
  - SystemVerilog
  - Platform Architecture
- **RTL**
  - VCS
  - NCsim
  - Modelsim, Vsim
  - JasperGold Formal
  - Verification
  - Verdi/nWave
- **Synthesis**
  - Genus
  - Design Compiler
  - Prime-Time
  - Tempus Timing Signoff Solution
- **Physical Design**
  - INNOVUS
  - IC Compiler
  - Calibre
  - Nanosim, CustomSim
  - Spectre APS/XPS, UltraSim

N16 ADFP Cell-based Design Flow

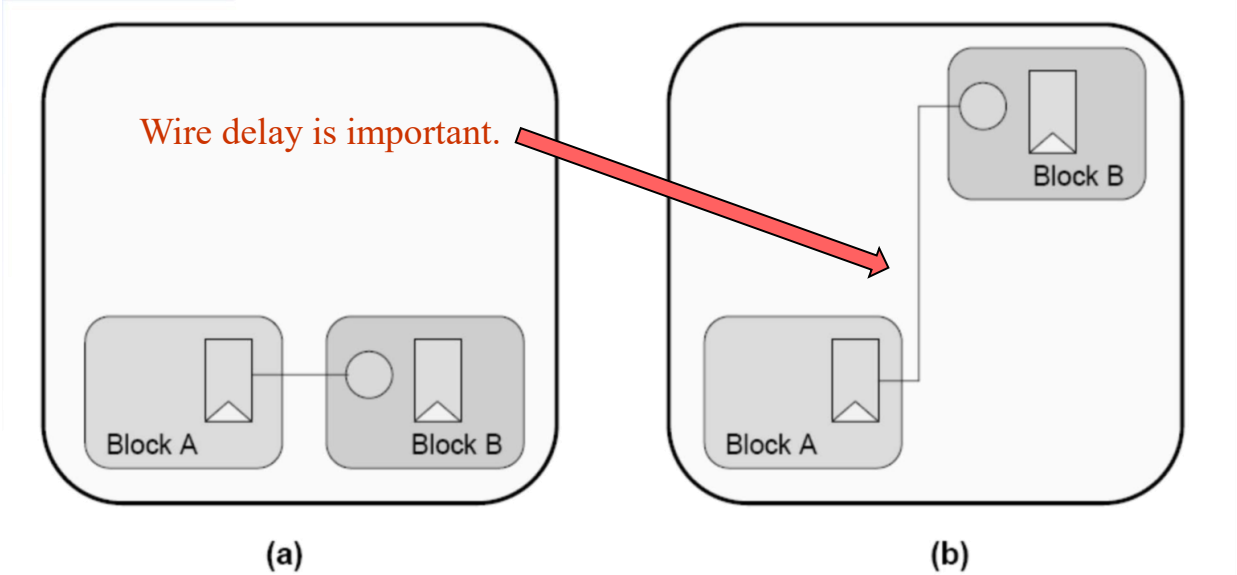
- Cell-based IC Design Flow



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Wiring Problem

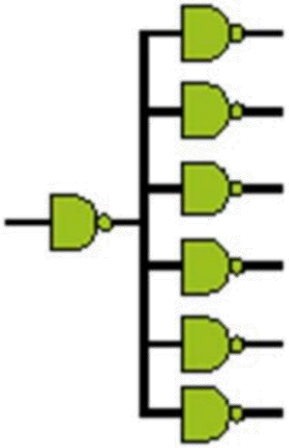
- **Floorplan affects timing budget and wire-load model.**
  - Wire delay can only be estimated in synthesis level.
  - Wire delay can be calculated after APR and RC extraction.



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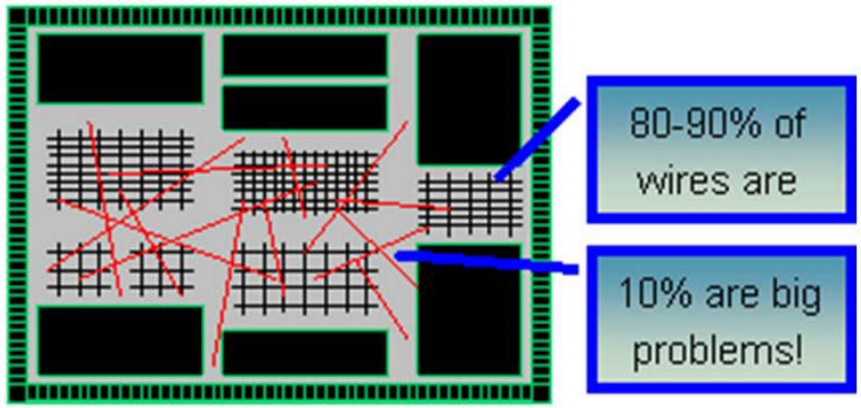
Wiring Problem (cont'd)

- **Difference in Synthesis View and Physical View**



Synthesis View

All wires of fan-out=n are the same.

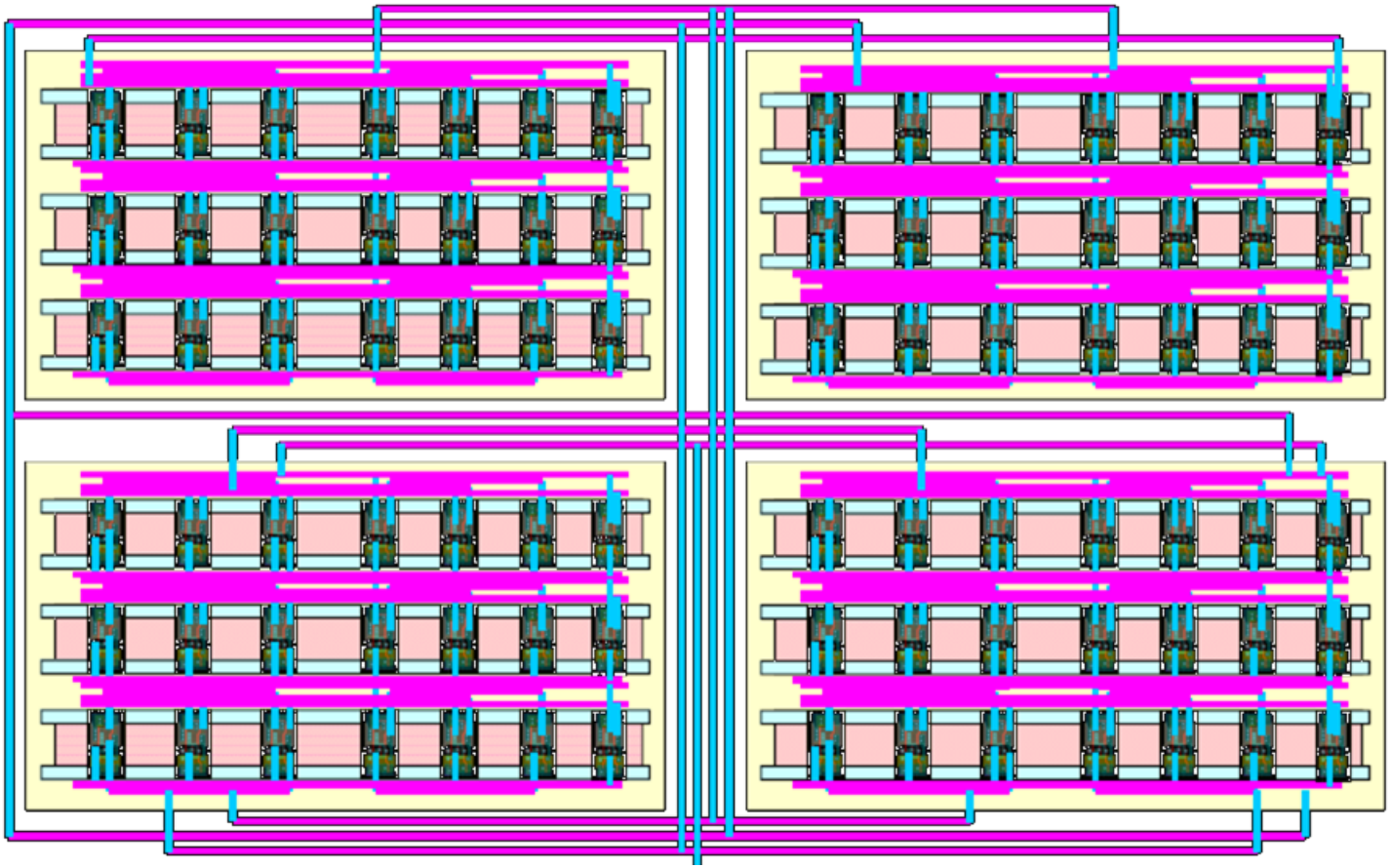


Physical View

Each wire is unique.

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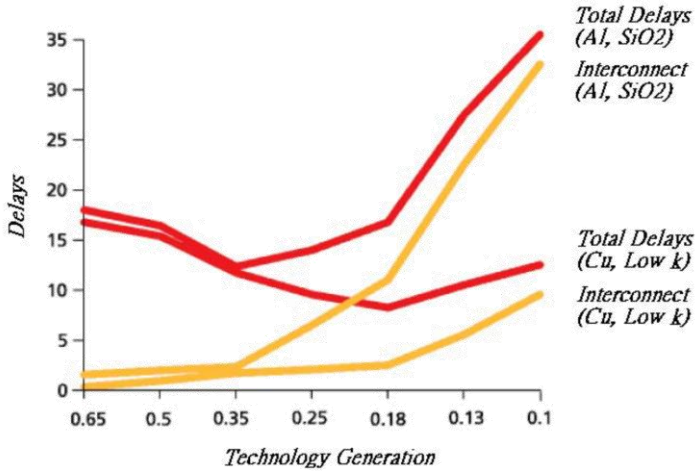
Wiring Problem (cont'd)

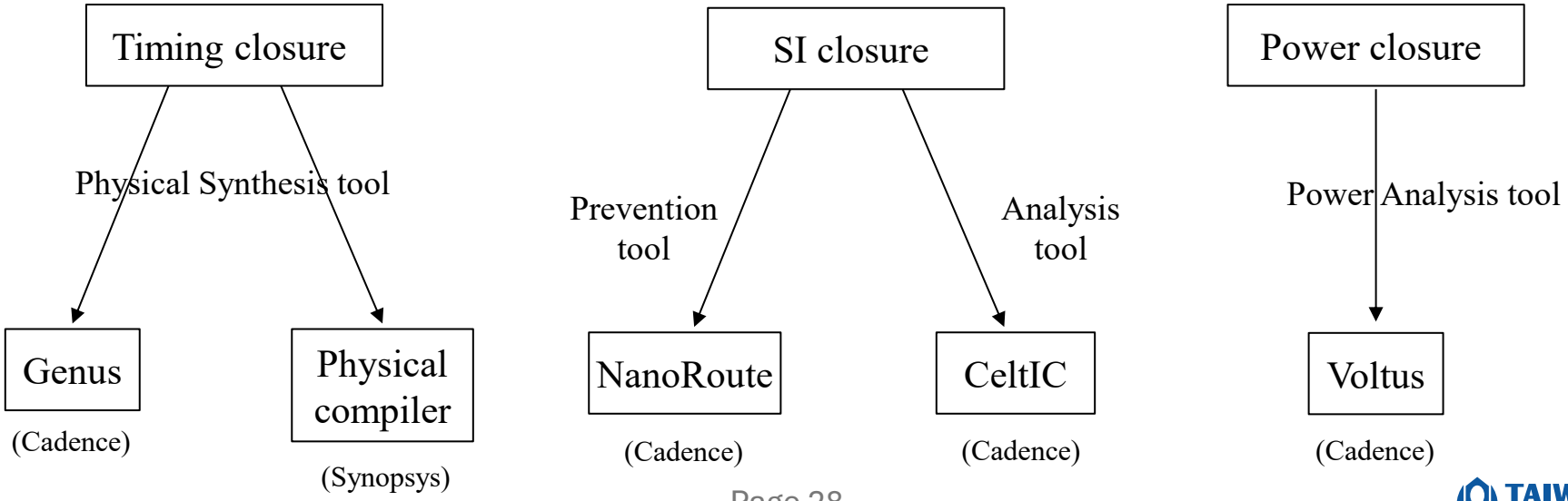


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Wiring Problem (cont'd)

- **Wiring delay dominates overall delay**
- **New problems due to large wire resistance**
  - Timing closure
  - Signal Integrity closure (crosstalk, …)
  - Power closure (IR drop, …)

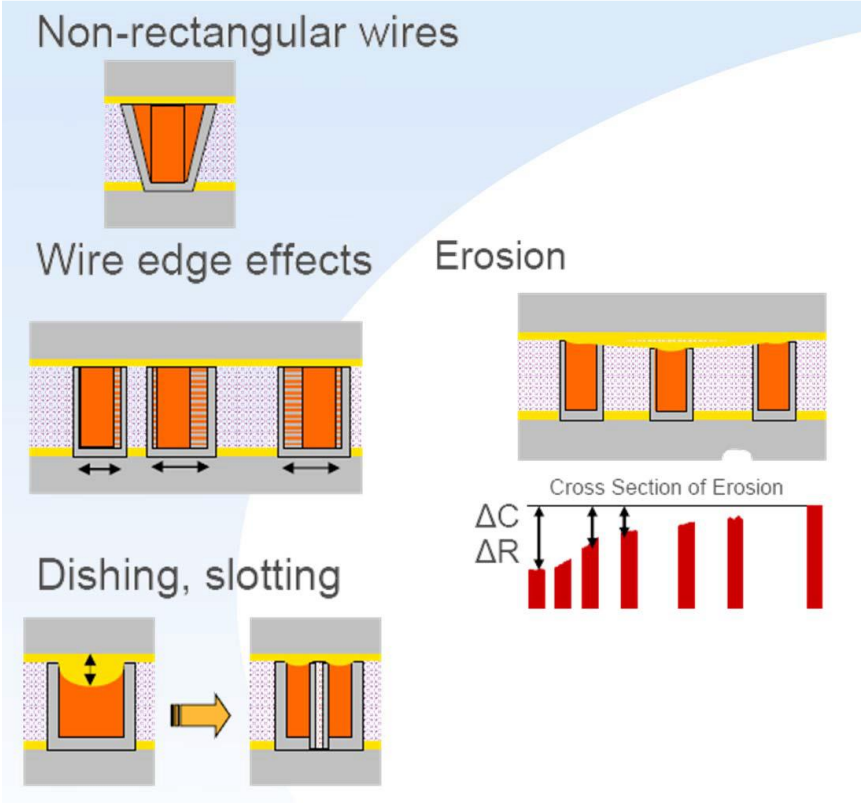


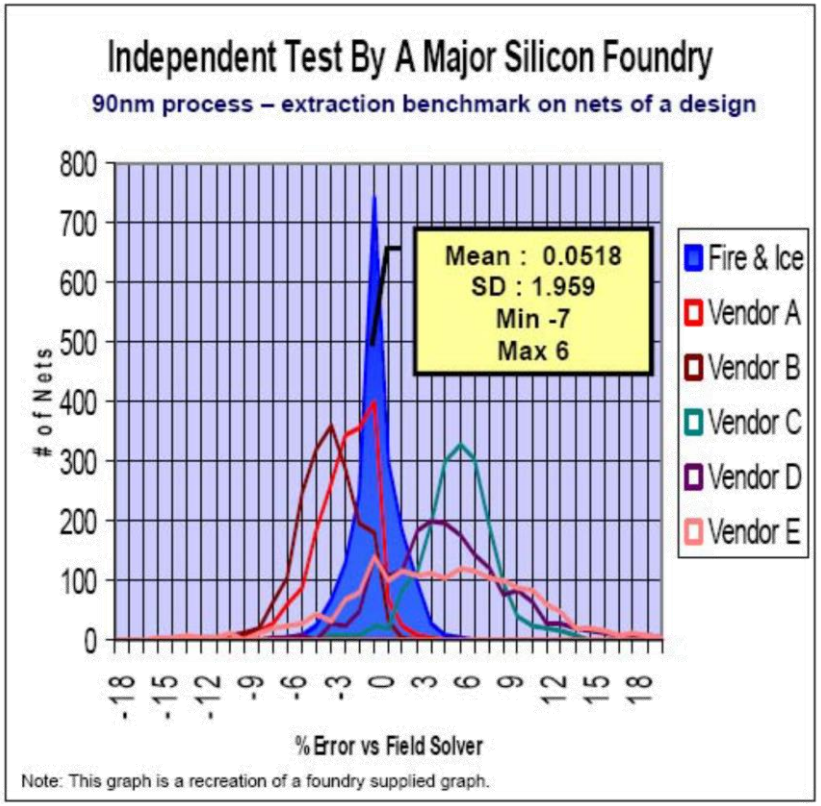


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Gate-Level RC Extractor

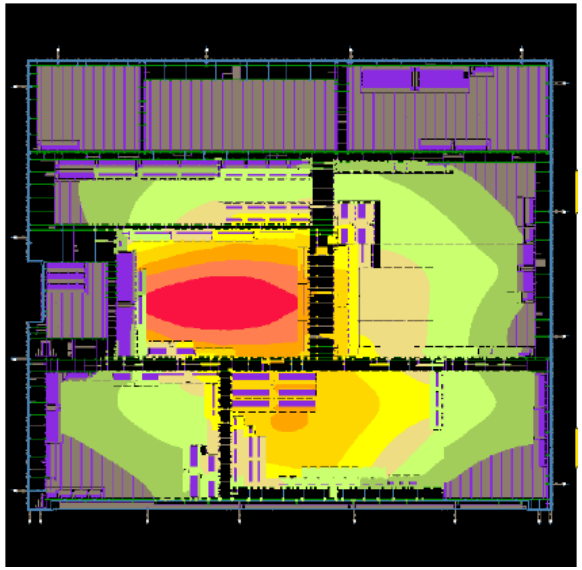
- QRC for for 3D parasitic extraction

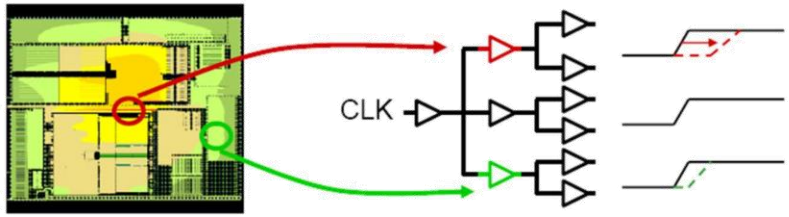


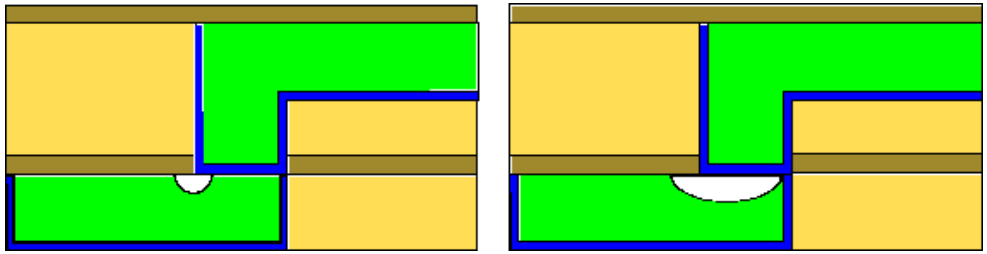


Power Grid Analysis

- IR drop and Ground bounce combine to impact silicon performance (setup/hold time violations)
- Electromigration (EM) caused by current flow and current density may make chip failure

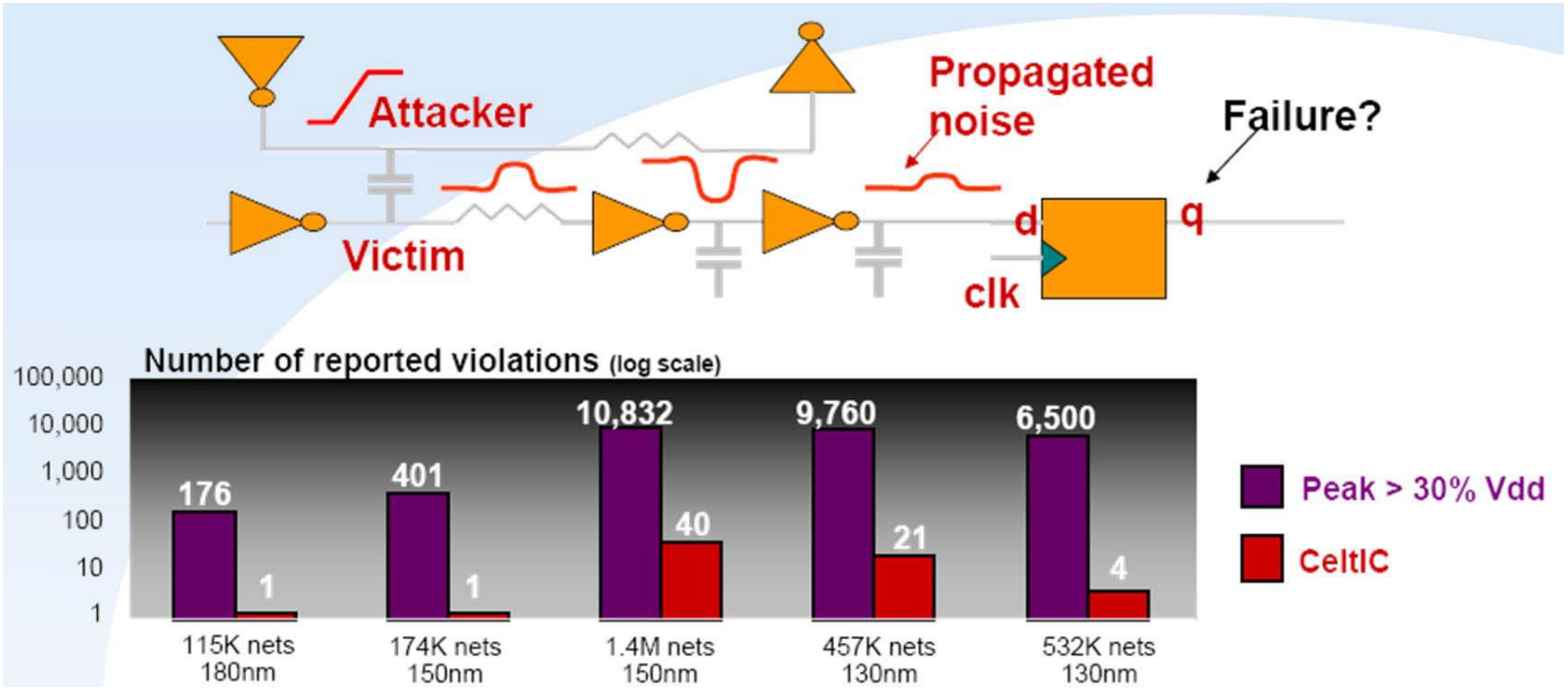






Glitch Analysis

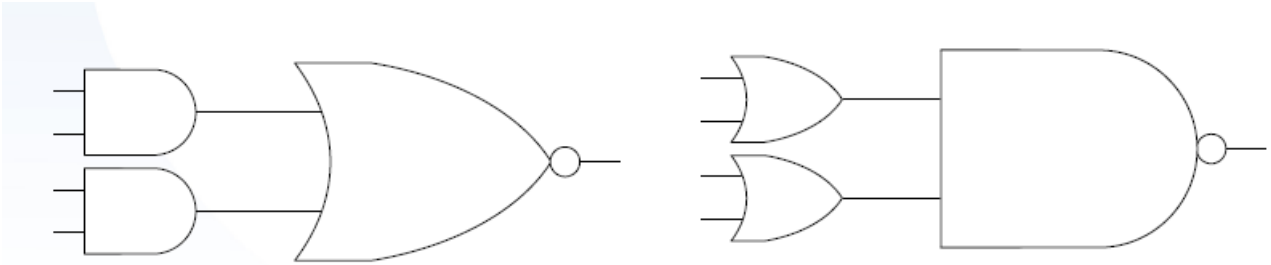
- Crosstalk glitches propagate to latches to create functional failures.
- Glitch Analysis is needed to detect crosstalk induced functional and timing failures
- Spice-like accuracy using mixed cell and transistor modeling



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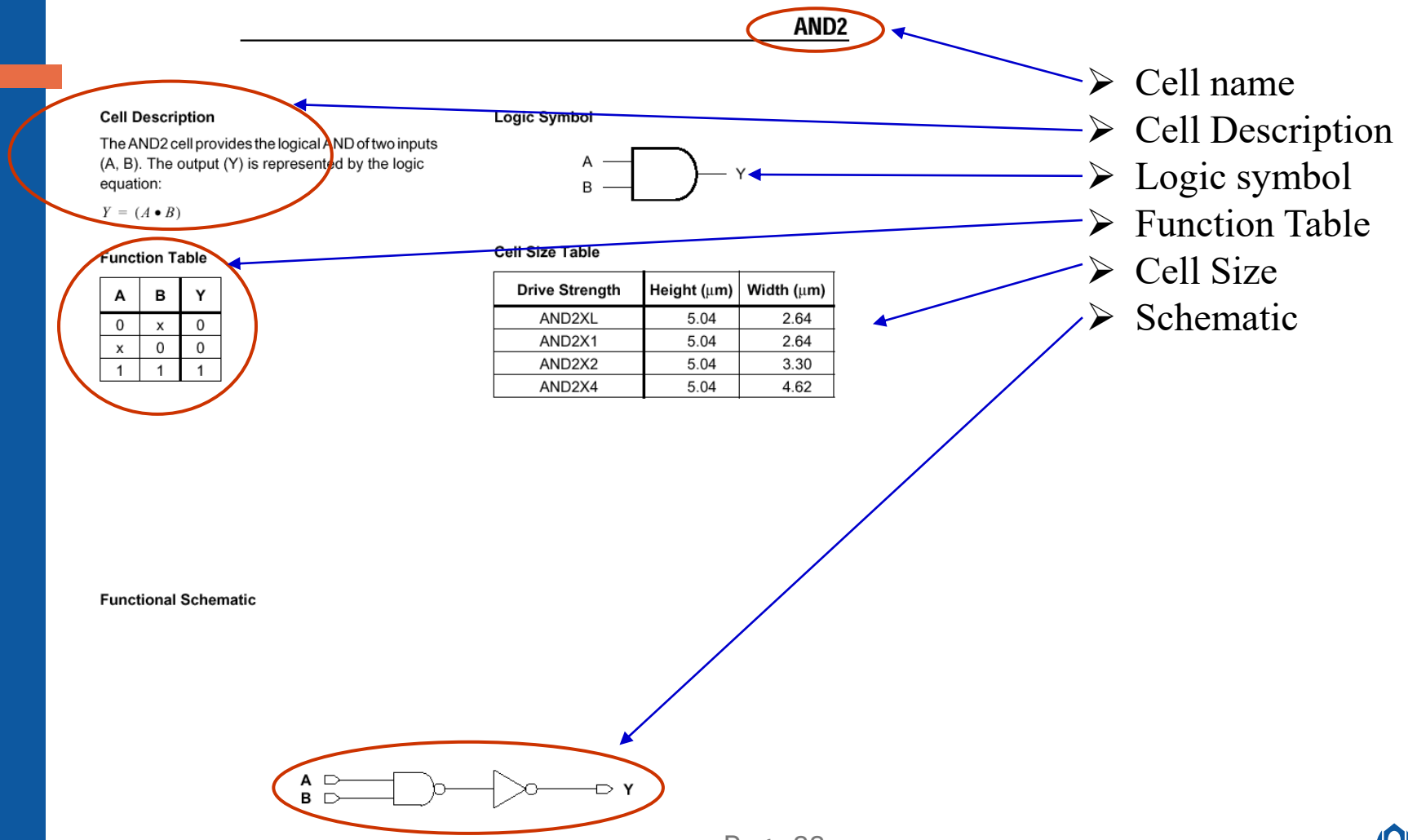
Introduction to Standard Cell Library

- **Standard Cell Library includes varies type of cells with different driving strength for optimization.**
- **Cell Type:**
  - Combinational: AND, OR, AOI, OAI, CLKBUF, TBUF,…
  - Sequential: DFF, DFF with Set/Reset, Scan DFF, Latch…
  - I/O Cell: Input/Output PAD, Power PAD……
  - Special: Antenna-Fix Cell, Fill Cell, Delay Cell…..
- **RAM/ROM Compiler and Register File Compiler:**
  - Generate required single/dual port memory and register files



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Cell Library: Data Sheet



- Cell name
- Cell Description
- Logic symbol
- Function Table
- Cell Size
- Schematic

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Cell Library: Data Sheet (cont'd)

~~AND2~~

AC Power Table

Pin	Power (μW/MHz)			
	XL	X1	X2	X4
A	0.0173	0.0192	0.0304	0.0540
B	0.0197	0.0215	0.0344	0.0610

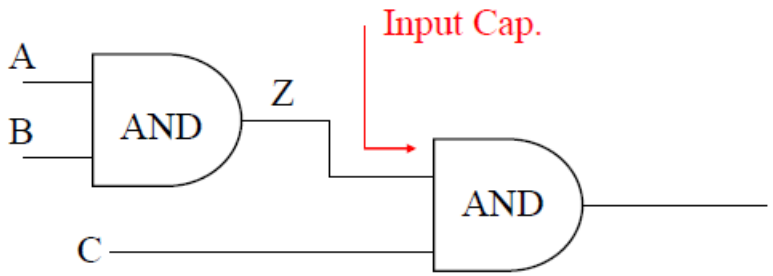
Pin Capacitance Table

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0021	0.0020	0.0034	0.0054
B	0.0023	0.0021	0.0034	0.0059

Delays Table at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A → Y↑	0.085	0.089	0.079	0.076	5.792	4.106	2.056	1.028
A → Y↓	0.096	0.118	0.095	0.096	3.444	2.499	1.286	0.660
B → Y↑	0.093	0.097	0.084	0.081	5.792	4.104	2.056	1.028
B → Y↓	0.107	0.131	0.105	0.108	3.451	2.503	1.287	0.661

- AC power
  - The power consumption when cell transition
- Pin Capacitance
  - Input pin's capacitance
- Delays Model
  - Intrinsic delay : delay with no load
  - Delay = Intrinsic delay+Kload * Cload



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Cell Library: Delay Calculation

- Delay = Intrinsic delay + K_{load} * C_{load}
- C_{load} is the **total** input capacitance of the next stage!!

AND2

AC Power Table

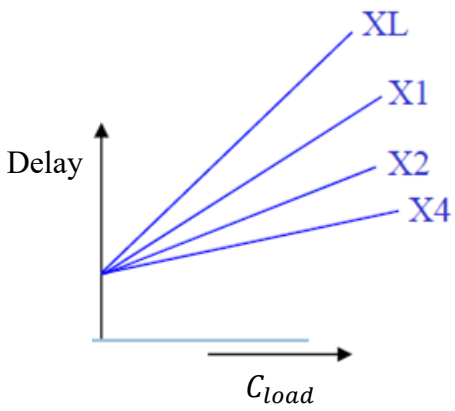
Pin	Power (μW/MHz)			
	XL	X1	X2	X4
A	0.0173	0.0192	0.0304	0.0540
B	0.0197	0.0215	0.0344	0.0610

Pin Capacitance Table

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0021	0.0020	0.0034	0.0054
B	0.0023	0.0021	0.0034	0.0059

Delays Table at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A → Y↑	0.085	0.089	0.079	0.076	5.792	4.106	2.056	1.028
A → Y↓	0.096	0.118	0.095	0.096	3.444	2.499	1.286	0.660
B → Y↑	0.093	0.097	0.084	0.081	5.792	4.104	2.056	1.028
B → Y↓	0.107	0.131	0.105	0.108	3.451	2.503	1.287	0.661



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Cell Library: Power Calculation

- Power = $F_{in} * P_{cell} + C_{load}\, VDD^2 * F_{out}$
- F_{in} is the change rate of input state
- P_{cell} is cell's AC power (μW/MHz)
- C_{load} is the **total** input capacitance of the next stage!!
- F_{out} is output state change rate

AND2

AC Power Table

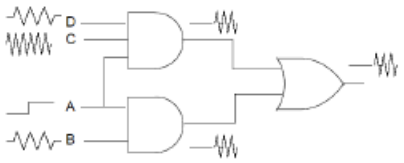
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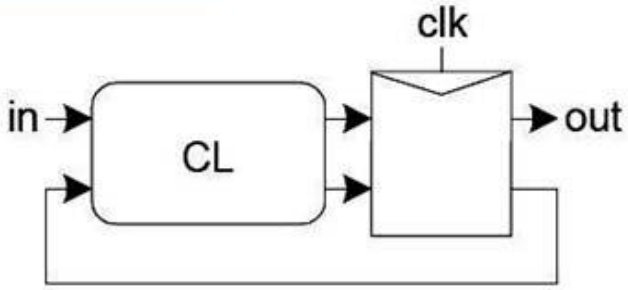
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	XL	X1	X2	X4	XL	X1	X2	X4
A → Y↑	0.085	0.089	0.079	0.076	5.792	4.106	2.056	1.028
A → Y↓	0.096	0.118	0.095	0.096	3.444	2.499	1.286	0.660
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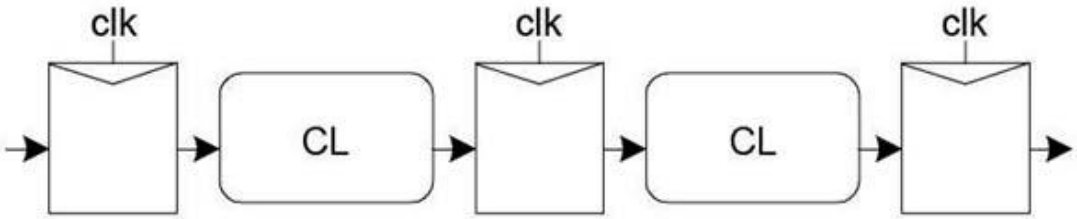
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Sequential Machine

- **A sequential machine is a machine**
  - for which the output values depend **not only on the present input values** but also the history of previous input.
  - Memory elements are added to combinational logic to build sequential machines.



Finite State Machine

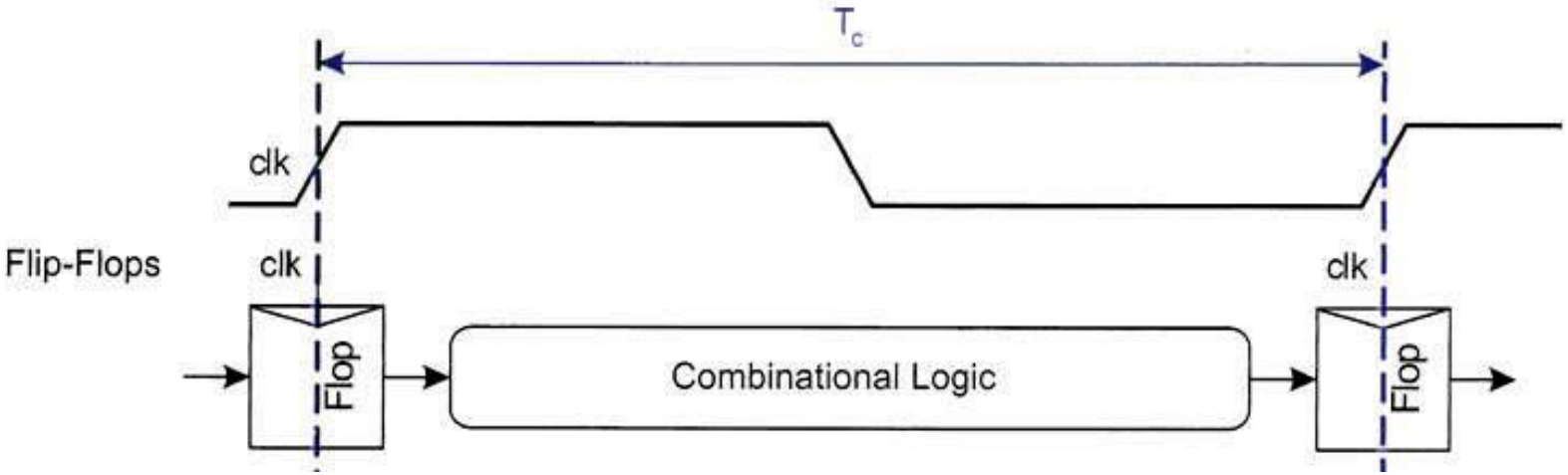


Pipeline

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Sequencing Methods

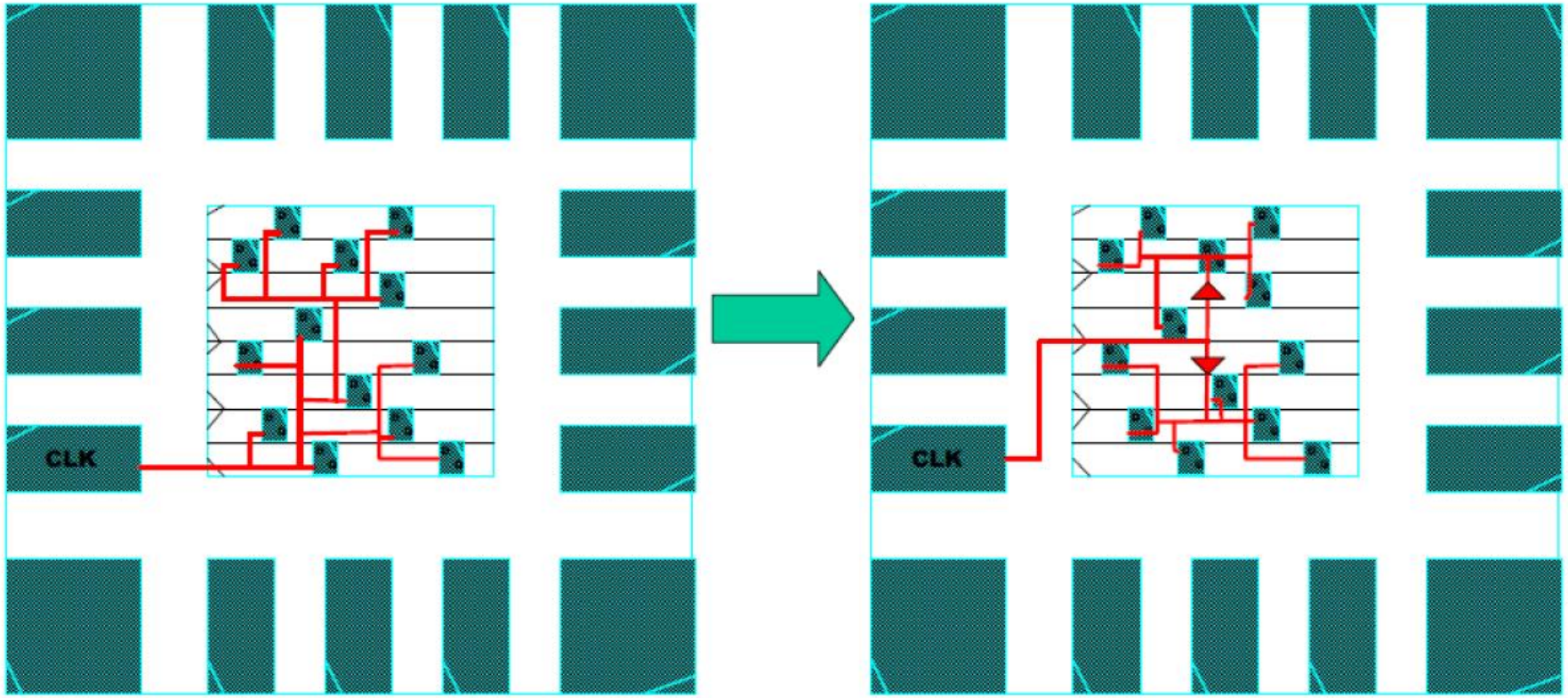
- **The sequencing blocks of combinational logic.**
  - The token is captured in the first flip-flop on the first rising edge of the clock.
  - It propagates through the combinational logic and reaches the second flip-flop on the second rising edge of the clock.
  - The clock period is Tc.



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Clock Tree Synthesis (CTS)

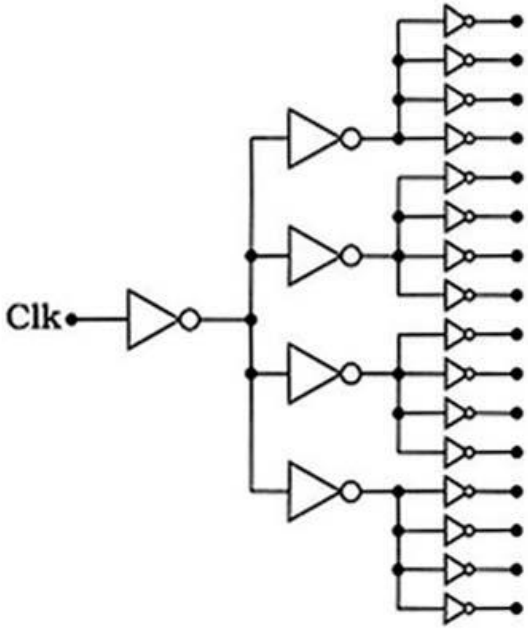
- **Clock is distributed to on-chip modules.**



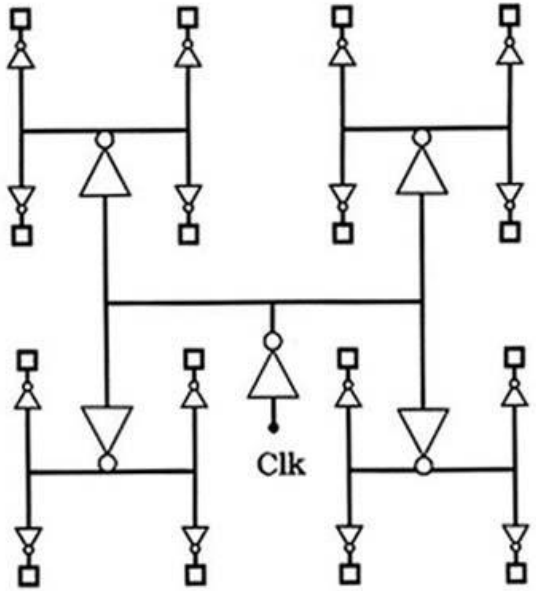
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Clock Driver Tree

- **The H-tree structure is created to have equal length for every clock path to minimize clock skew.**



Clock Driver

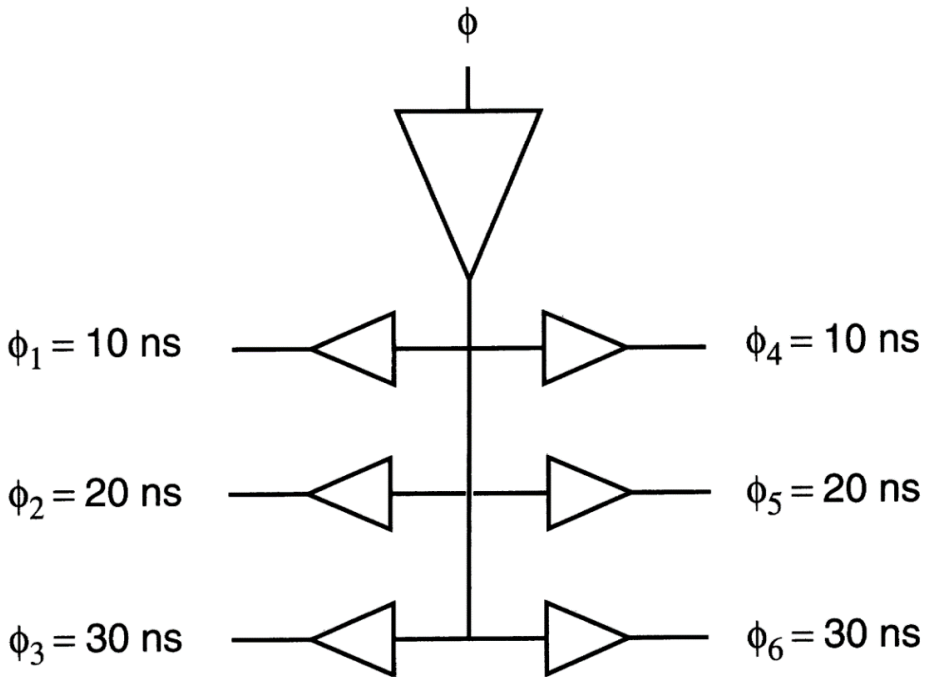


H-tree

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Clock Network Delay

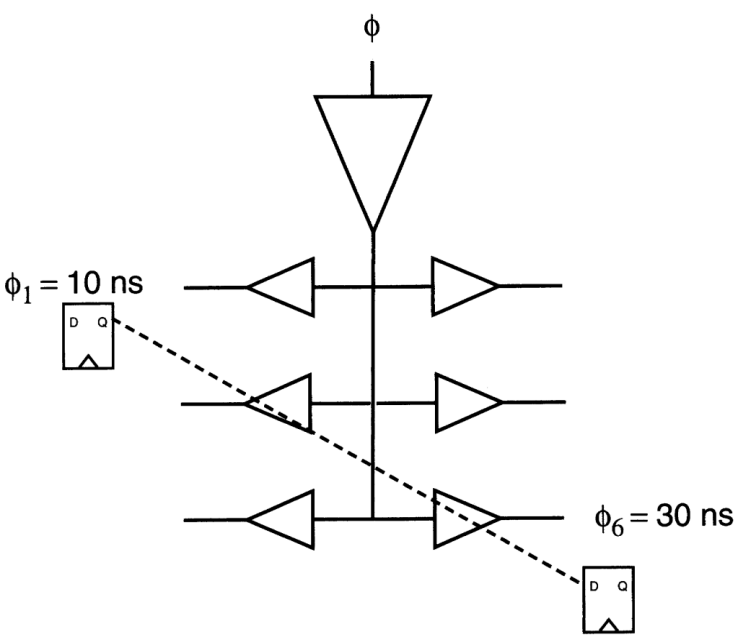
- **Because there are so many flip-flops on the chip, we need clock tree buffers to drive these flip-flops.**
  - The delay in this clock distribution network is shown.

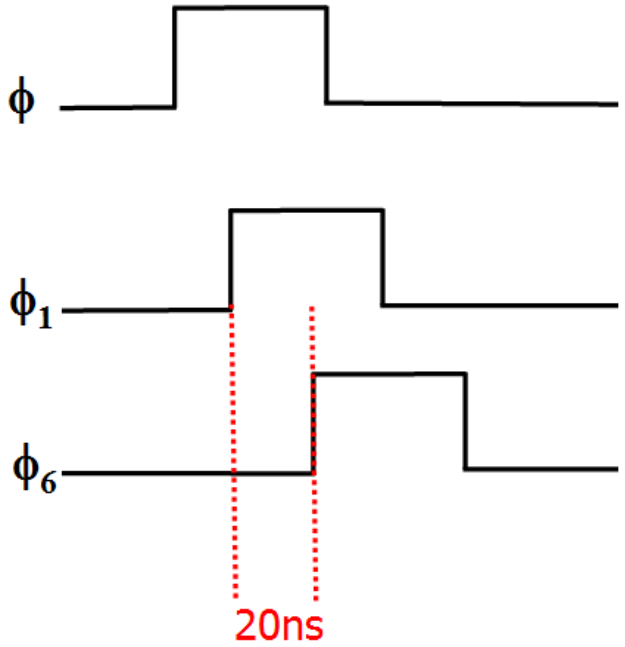


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Clock Skew

- **Clock skew describes a relative delay or offset in time between any two clock signals.**
  - Clock skew between φ1 and φ6 is 20ns.

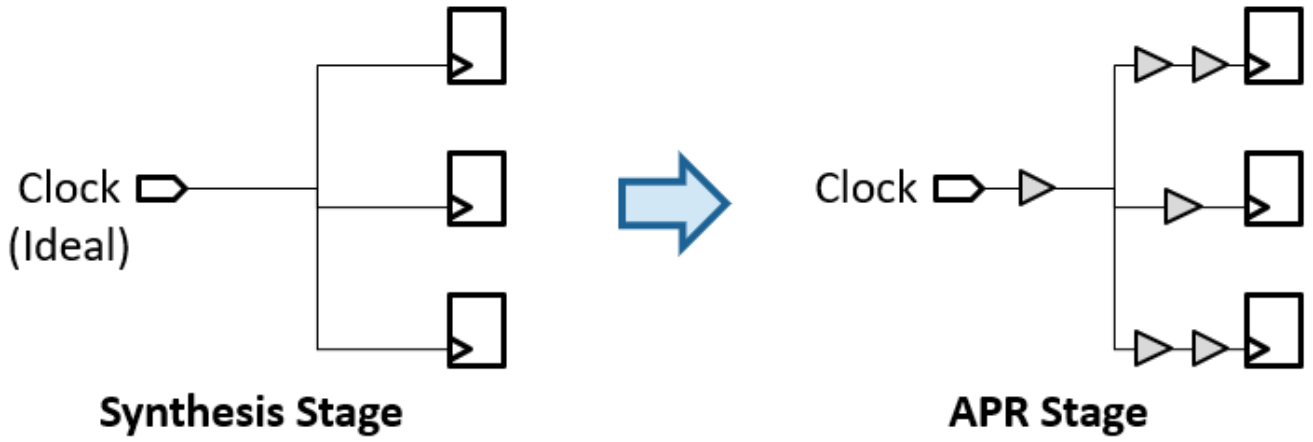




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Optimization Example of CTS

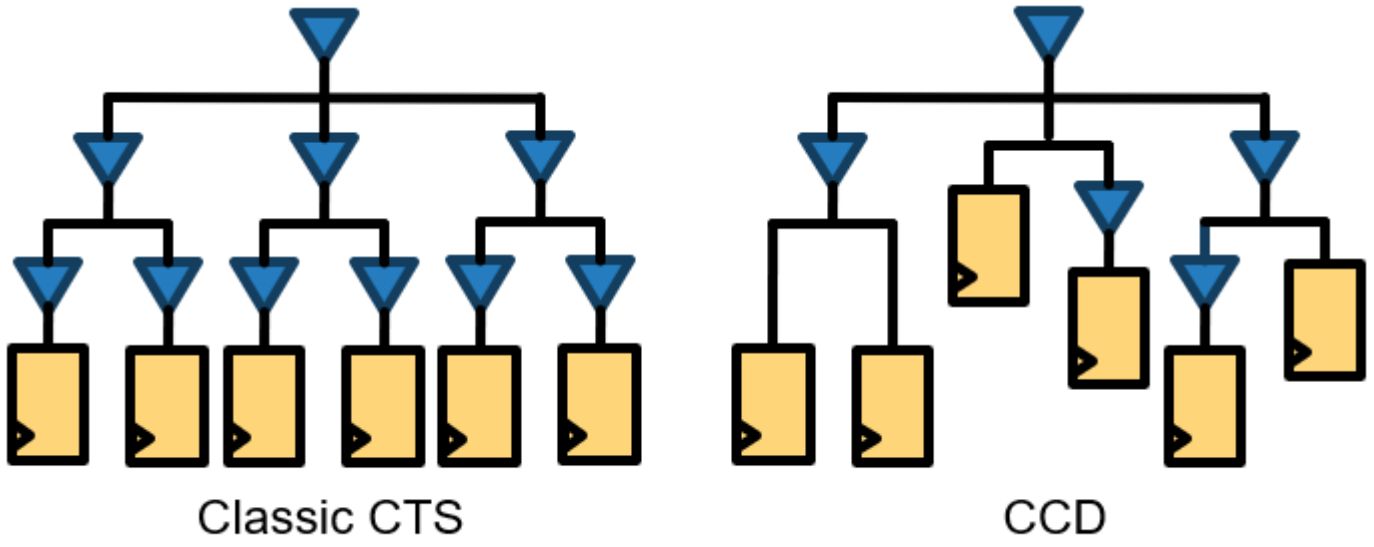
- **During the APR stage, CTS is performed to build the actual clk buffer / inverter tree, so the clack has real wire RC, and clk skew.**
- **Since a real clk tree cannot be constructed during the synthesis stage, an abstract model is used instead, which is referred as Clock Modeling**



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Concurrent Clock and Data Flow (CCD)

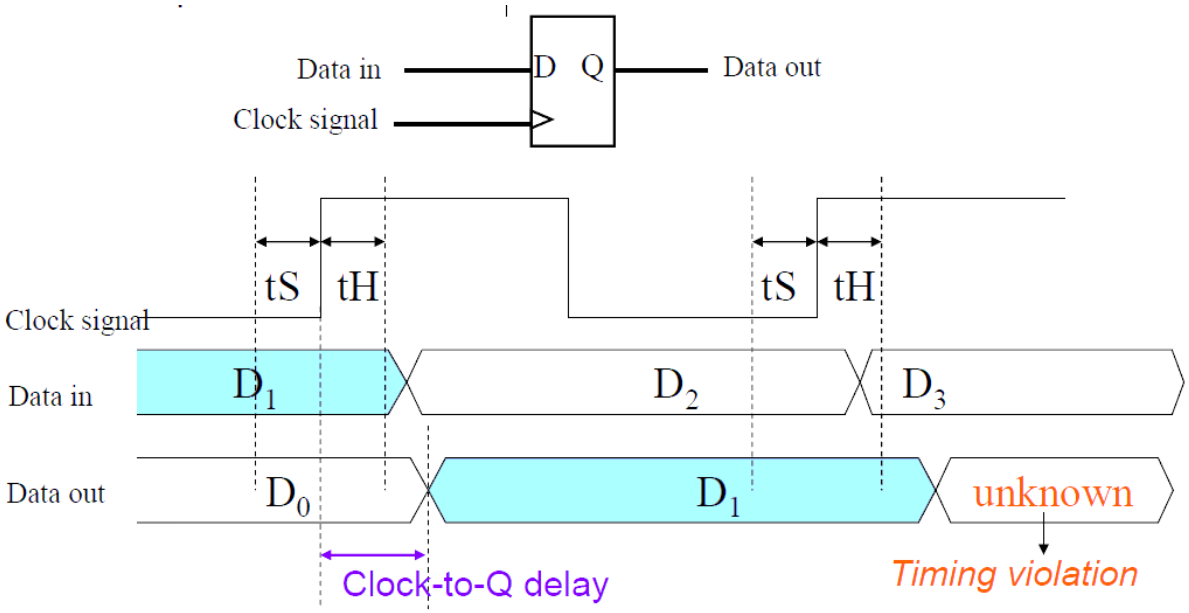
- CCD is used to optimize the clk network by
  - Reasonable use of skewing
  - Simultaneous optimization of clk path and logic path



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Sequential Element: D-Flip/Flop

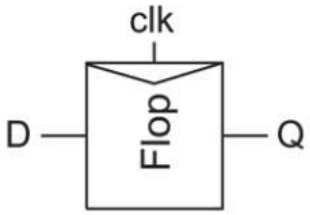
- **D-Flip/Flop Timing Violations**
  - Setup time and hold time

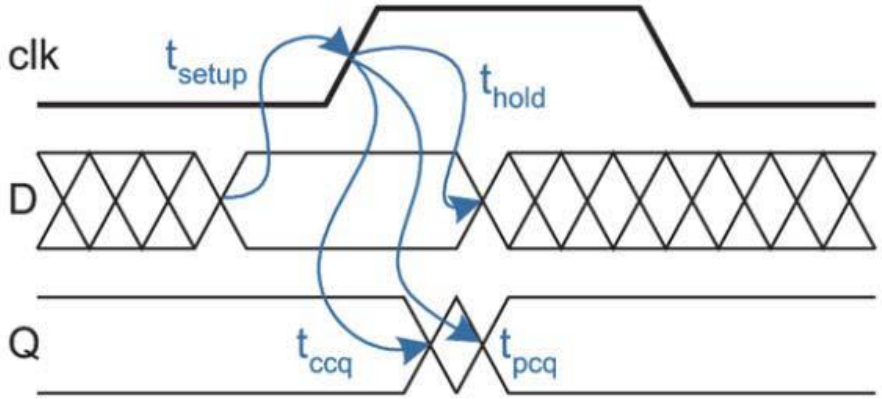


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Sequential Element: D-Flip/Flop (cont'd)

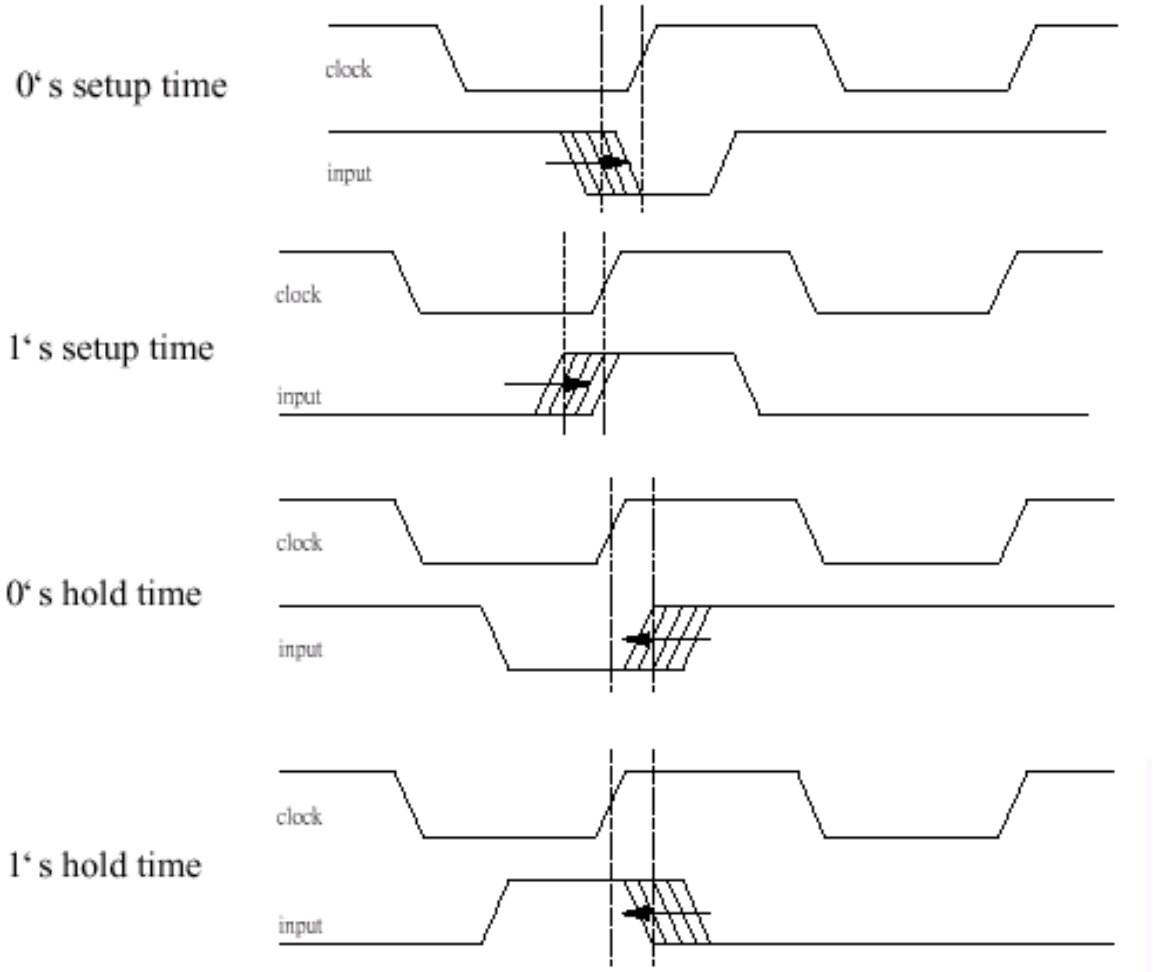
- **Timing definition**
  - Setup time (tS): The time that the input signal must be stabilized before the clock edge.
  - Hold time (tH): The time that the input signal must be stabilized after the clock edge.
  - Clk-to-Q delay (tQ): The propagation time that Q reaches its steady state after the clock edge





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Setup/Hold Time Measurement (cont'd)



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Setup Time and Hold Time

- **Setup Time**
  Data must arrive and stabilize at the capture flip-flop before the next active clock edge.
- **Maximum-delay Check**

$$T_{clk} + T_{skew} > T_{cq} + T_{comb(max)} + T_{setup}$$

- **Hold Time**
  New data must not reach the capture flip-flop too early after the same active clock edge.
- **Minimum-delay Check**

$$T_{cq} + T_{comb(min)} > T_{hold} + T_{skew}$$

T_{cq}: $\boldsymbol{Flop\ \ clock\ \ to\ \ Flop\ \ q\ \ delay}$,

T_{setup}/T_{hold} : $\boldsymbol{Required\ setup/hold\ \ time\ of\ a\ Flop}$

T_{skew} :$\boldsymbol{Delay\ between\ clock\ edges\ of\ two\ adjacent\ flops}$

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Setup Time Example

- Given the following timing parameters:

Parameter	Value
Clock period T_{clk}	10 ns
Clock-to-Q delay T_{cq}	1 ns
Maximum combinational delay $T_{comb(max)}$	7 ns
Setup time T_{setup}	1 ns
Clock skew T_{skew}	0.5 ns

Determine whether the setup time constraint is satisfied.
Calculate the setup slack.

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Setup Time Example

- Setup time constraint:

$$T_{clock} + T_{skew} > T_{cq} + T_{comb(max)} + T_{setup}$$

Substitute the values:

$$10 + 0.5 > 1 + 7 + 1 = 10.5 > 9$$

Setup Slack:

$$Slack_{setup} = 10.5 - 9$$

$$= \boxed{1.5 \text{ ns}}$$

So the setup time constraint is satisfied.

No setup violation occurs.

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Hold Time Example

- Given the following timing parameters:

Parameter	Value
Clock-to-Q delay T_{cq}	0.5 ns
Minimum combinational delay $T_{comb(min)}$	0.3 ns
Hold time T_{setup}	0.6 ns
Clock skew T_{skew}	0.4 ns

Determine whether the hold time constraint is satisfied. If a violation occurs, calculate the minimum additional delay required to fix it.

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Setup Time Example

Hold time constraint:

$$T_{cq} + T_{comb(min)} > T_{hold} + T_{skew}$$

Substitute the values:

$$0.5 + 0.3 > 0.6 + 0.4 = 0.8 > 1.0$$

Hold Slack:

$$Slack_{hold} = 0.8 - 1.0$$
$$= \boxed{-0.2 \text{ ns}}$$

So a hold time violation occurs.

At least 0.2 ns of additional delay is required to meet the hold constraint.

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Contents of Cell Library

- **Functional model (Verilog model, .v)**
  - for synthesis and simulation
- **Symbol (Schematic)**
  - for schematic entry circuit diagram display
- **Timing/Power model (.lib)**
  - for synthesis and simulation
- **Netlist (SPICE netlist, .sp)**
  - layout verification and transistor-level simulation

Contents of Cell Library (cont'd)

- **Layout abstract (Library Exchange Format, .lef)**
  - for Auto Placement and Routing (APR)
- **Noise model (Celtic database, .cdb)**
  - for cross talk analysis
- **Power network view model (Voltus database)**
  - For Static/Dynamic IR-drop analysis
- **Layout (GDSII, .gds)**
  - for final layout and tape-out
- **User guide and data sheet**

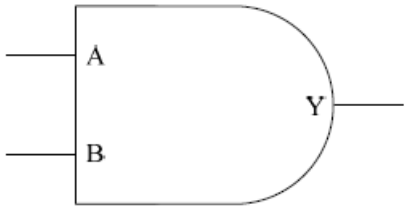
Major Considerations for Whole Cell Library

- **Area**
  - not only small cell size, but also routability
- **Timing**
  - acceptable range of device size, provide more flexible driving classes
- **Power**
  - Small device size as possible
  - Multiple VDD
  - Multiple Vt

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Design Considerations : Simulation

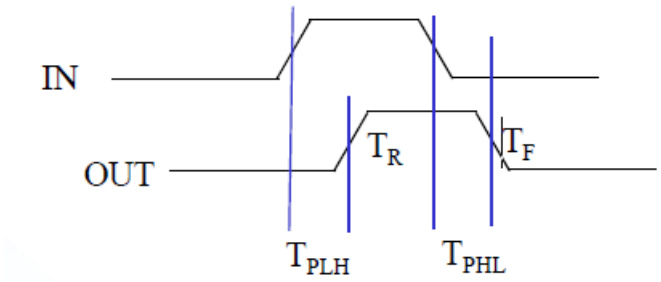
- **Full coverage process, voltage, and temperature variations (PVT Variations)**
  - Process: FF, TT, SS
  - Voltage: 1.1*VDD, VDD, 0.9*VDD
  - Temperature: -40˚C, 25˚C, 125˚C
    - Best-case: (FF, 1.1*VDD, -40˚C)
    - Typical-case: (TT, VDD, 25˚C)
    - Worst-case: (SS, 0.9*VDD, 125˚C)
- **Full coverage for input signal combination**
  - Ex. N-input cell should be simulated 2N patterns
  - Both delay and power is state dependent.



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Design Considerations : Timing

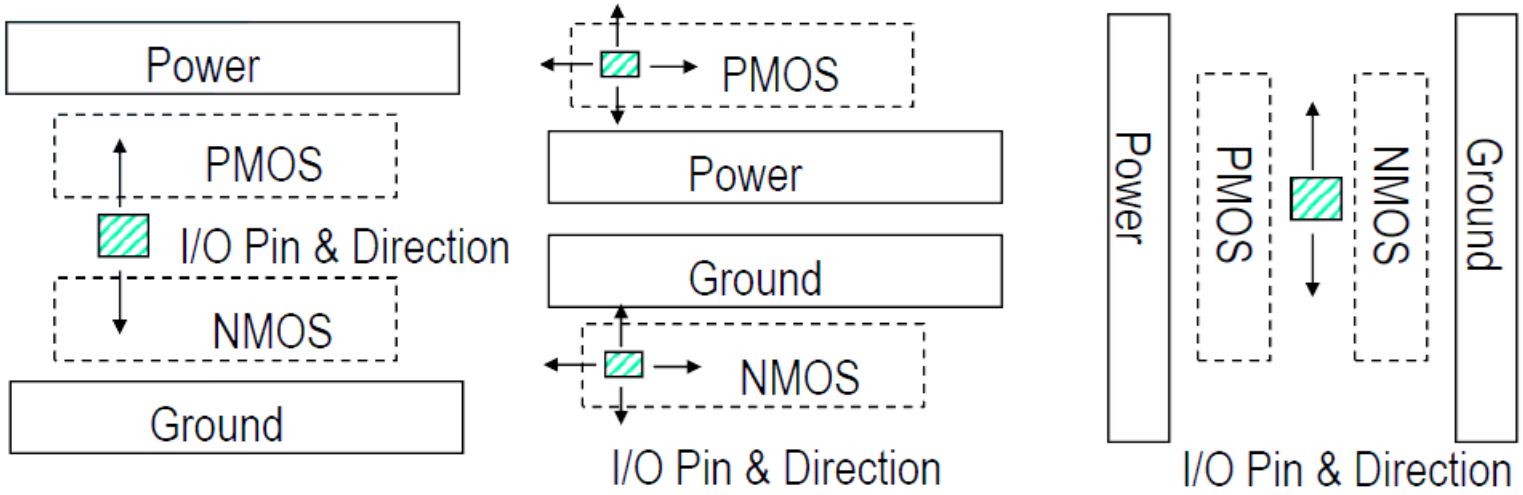
- **Four important parameters for general cells**
  - $t_{PLH}, t_{PHL}, t_R, t_F$
  - $t_{PLH} = t_{PHL}$ and $t_R = t_F$ should be better
  - t_R, t_F are more important than t_{PLH} and t_{PHL} on the CAD issue
  - These parameters must be measured over **different input skew** rate and **loading** to generate the final timing model
- **Four important parameters for sequential cells**
  - Setup, hold, release time and minimum pulse width
  - Zero or less than zero hold time will be better



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Cell Layout: Structure

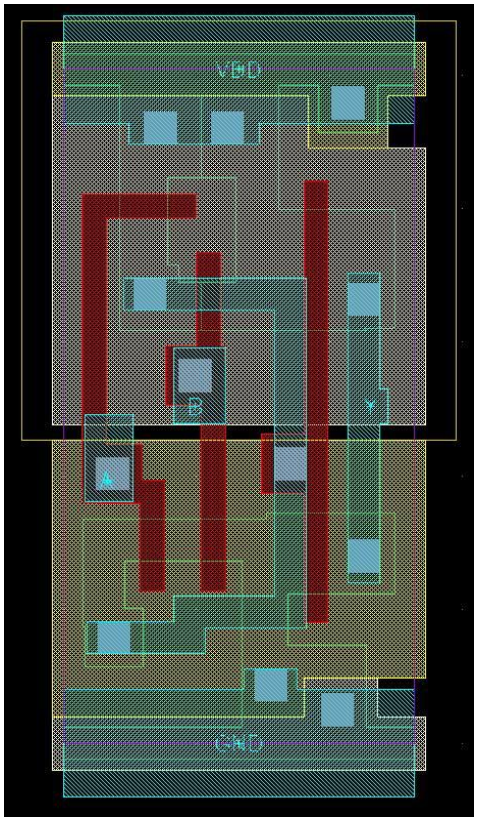
- **Width of Power /Ground Line & Location**
- **Placement of PMOS & NMOS**
- **Pin Size & Direction & Position**



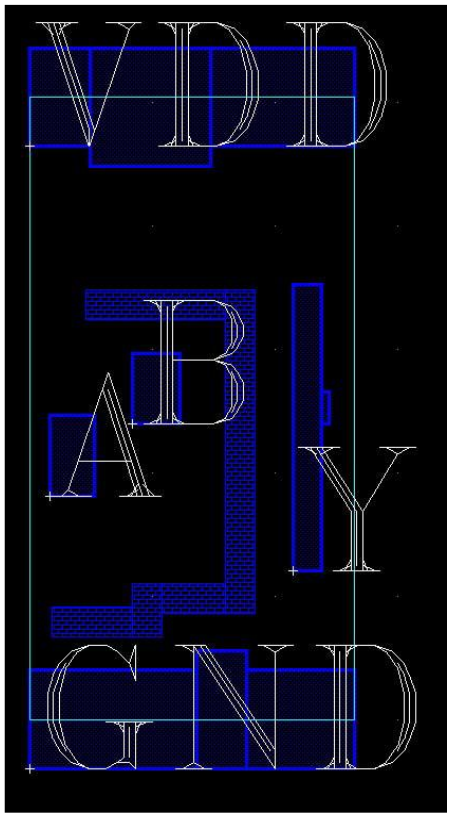
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Cell Layout and Abstraction

- **Two-input AND**



Layout View

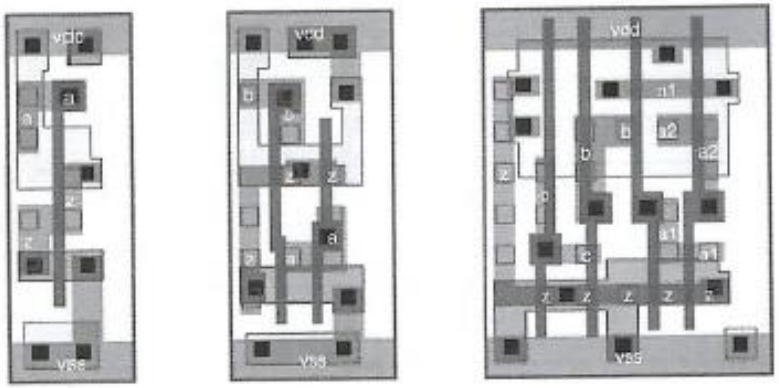


Abstract View

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Standard Cell Library

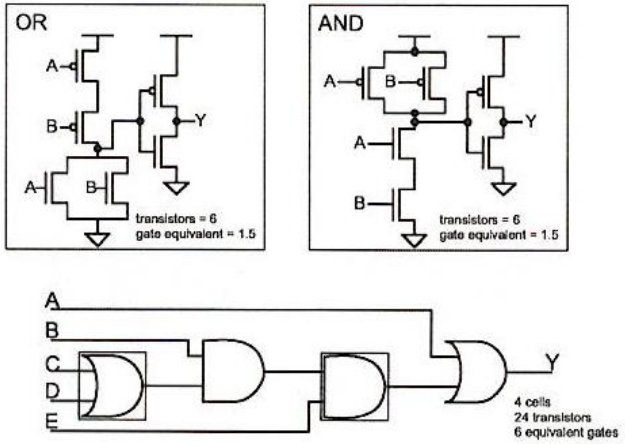
- Two types of cell:
  - Core cells
  - I/O cells (pads)
- Contains for each cell:
  - Functional information
  - Timing information
  - Power information
  - Layout (abstract) information
- Wire-load models



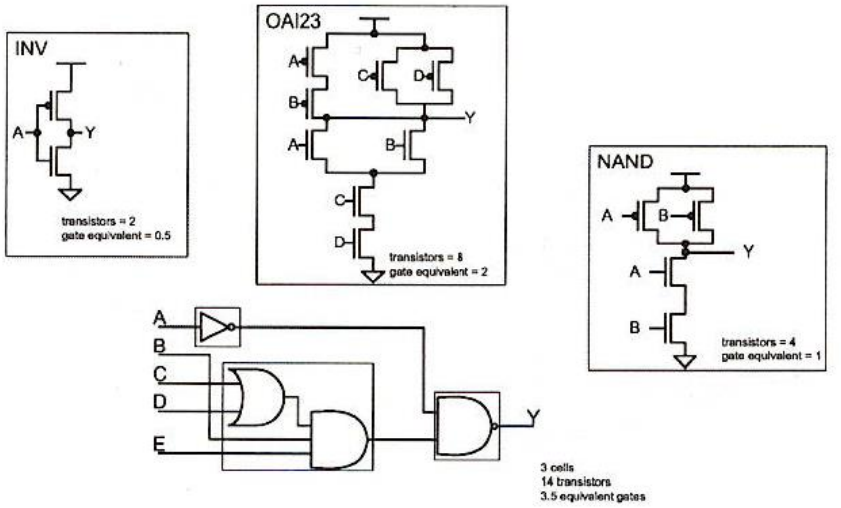
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Technology Mapping Example

Before optimization



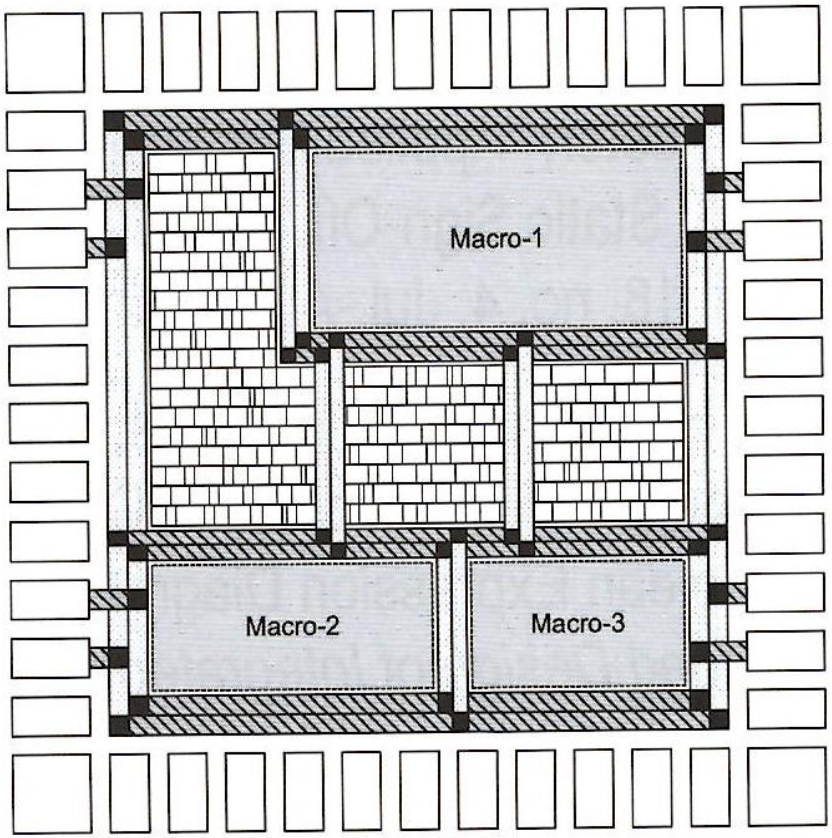
After optimization



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Cell-based Physical Layout

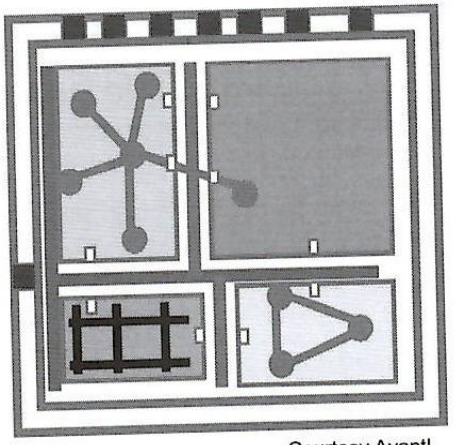
- **I/O Pads**
- **Hard Macros**
- **Standard Cells**
  - Flattening
  - Collect to soft blocks
  - Collect to groups & regions



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Physical Floorplanning

- **Soft macros , Hard macros , IP's and floorplan groups**
- **Block placement**
- **Interactive floorplan adjustment & edit**
  - Move , mirror , rotate , align
  - Change size , aspect ratio
  - Reserve routing layers
  - Add /delete/modifymove/align pins/bus pins
- **Pin assignment**
- **Early analysis : timing, congestion , …**

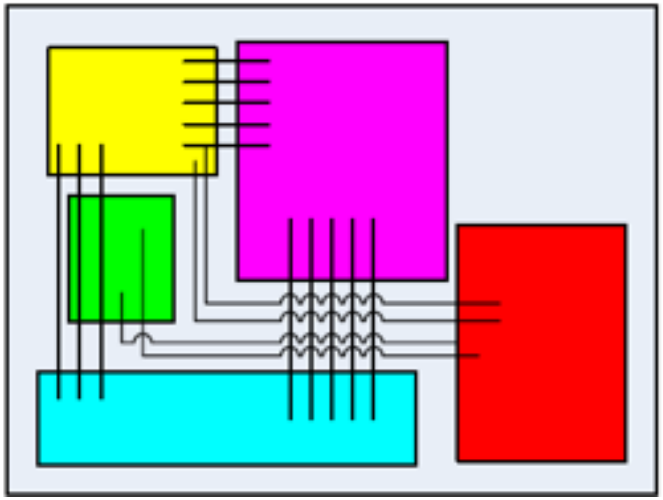


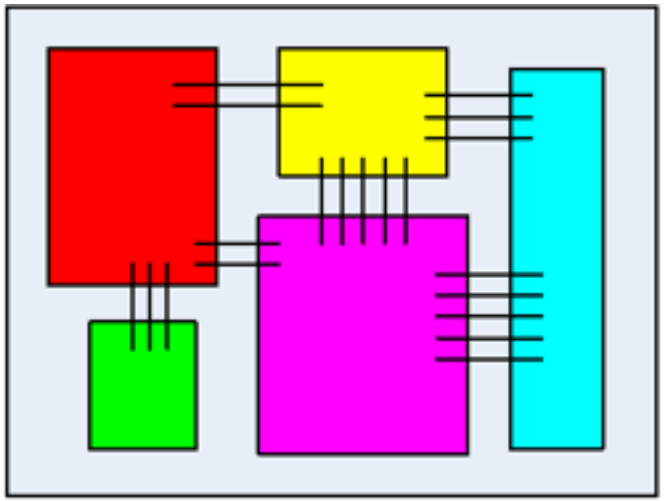
Courtesy Avant!

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Floorplan Issues

- **Issues**
  - **Congestion & Routing**
  - **Insertion POC (Power-On Control)**
  - **Power Planning**
  - **IR Drop**
  - **Timing**
  - **Utilization**





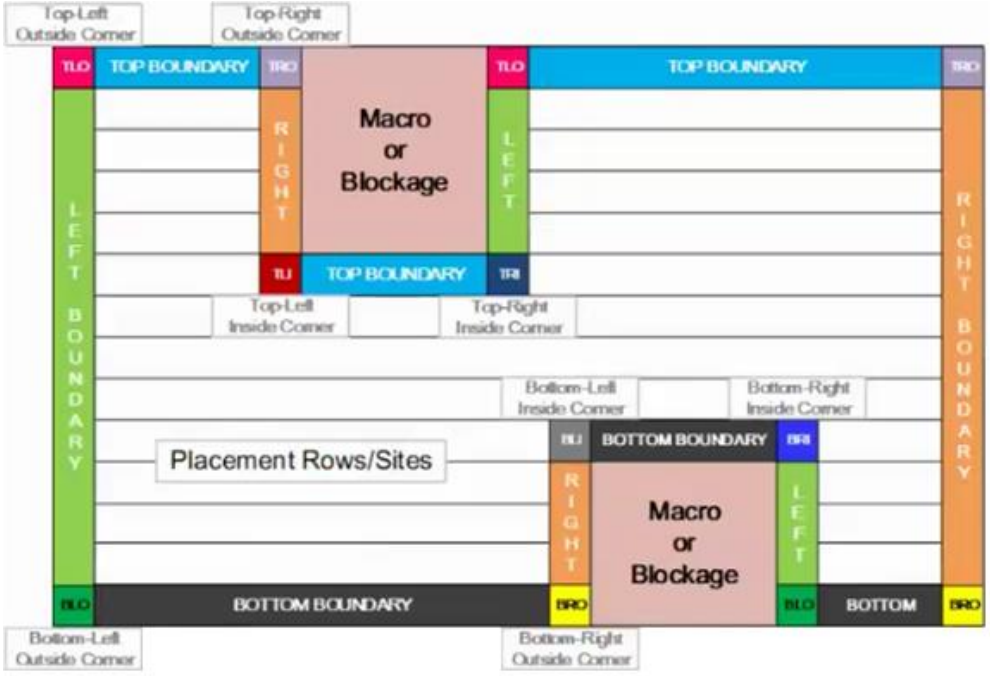
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Place Endcap Cells

- **Issues**

In the N16 process, compared to previous technologies, an additional endcap technique is introduced.

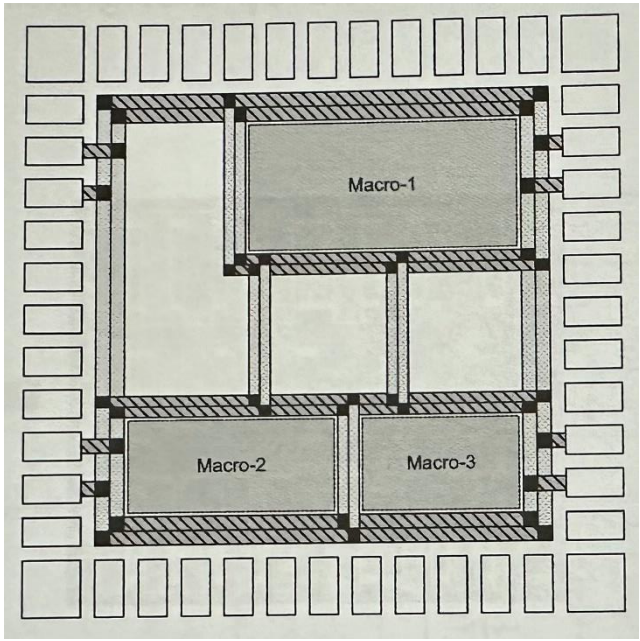
Before placing other cells, endcap cells are instead at both ends of each cell row to reduce timing variability at the core boundary cells.



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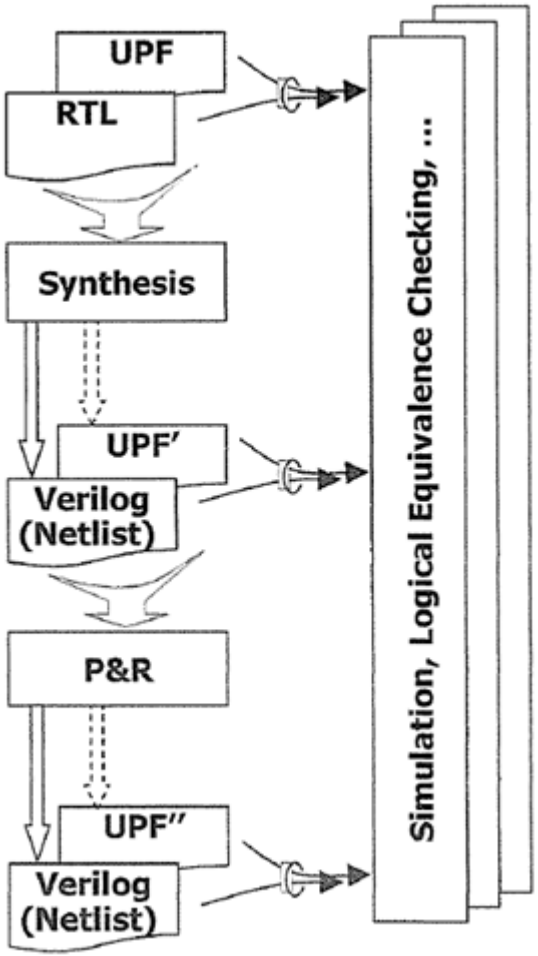
Power Frame

- Route power rings
- Route power straps
- Connect macros & pads to rings & straps



UPF Design Data Flow

- **UPF (Unified Power Format)**
  - **Initial UPF: High-Level Power Intent**
  - **UPF': Elaborated UPF from Synthesis**
  - **UPF'': Final UPF from Place & Route**



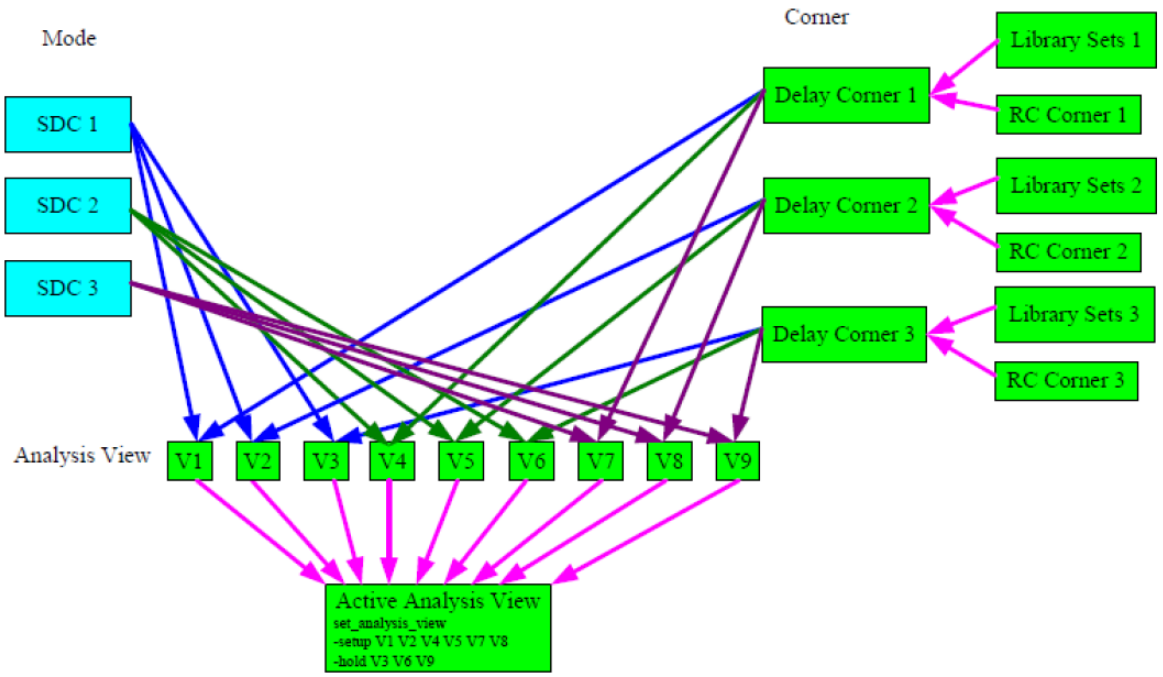
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MMMC Configuration

- **Multi-Mode Multi Corner (MMMC)**
  - Load timing information and .lib/.cdb file for signal integrity (SI) analysis (library setting)
  - RC Corners for extracting capacitive parameters
  - Operating conditions
  - Delay corners
  - Add constraint mode and read .sdc file
  - Create MMMC analysis views, assign which views are used for setup / hold time analysis and optimization

MMMC Example

- **Multi-Mode Multi Corner (MMMC)**
- Example: Design is required to meet 3 operating corners
  - Corner1: 0.81V, 125˚C
  - Corner2: 0.9V, 25˚C
  - Corner3: 0.99V, -40˚C
-



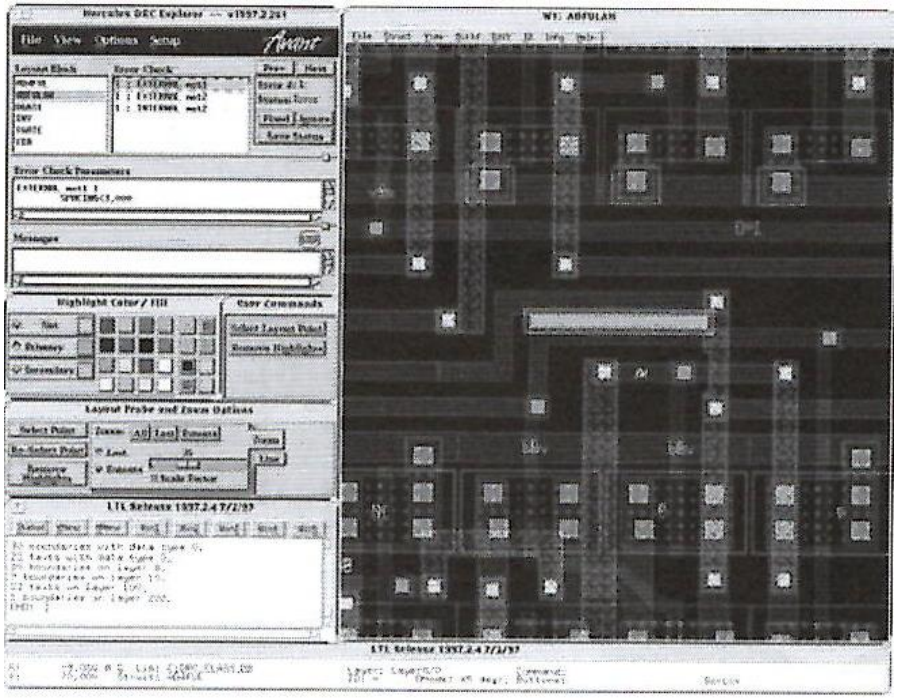
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Post-layout Verifications

- **For geometrical verification**
  - DRC , ERC
- **For topological verification**
  - LVS
- **For functional & performance verification**
  - Circuit extraction, RC extraction
  - Circuit-level post-sim / STA
  - Gate-level post-sim / STA
  - Power analysis : static & dynamic

DRC & ERC

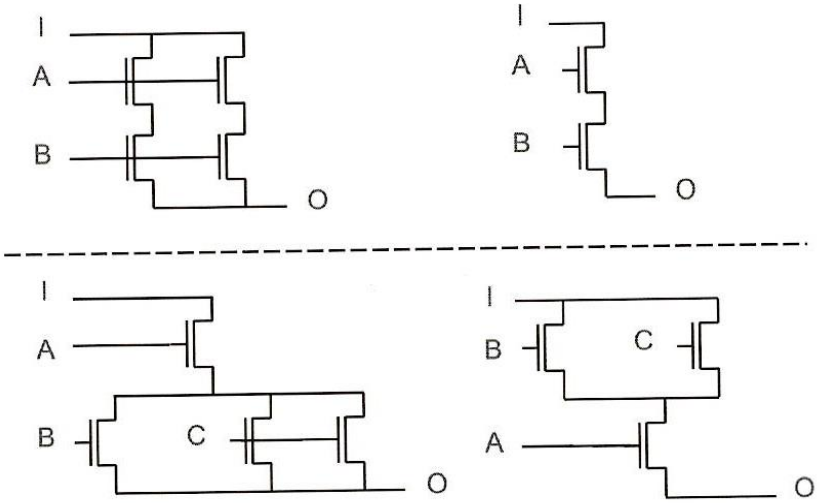
- Minimum width
- Minimum spacing
- Minimum enclosure
- Short
- Open
- Black box
- Whole chip



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LVS

- **Extract netlist from layout**
- **Perform device merging**
- **Compare netlists**



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